Prague University of Economics and Business
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A Decomposition of 10-K Risk-Factor Language and Post-Filing Stock Return Volatility

Bachelor Thesis

Study programme: Data Analytics

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Declaration on the Use of AI

I confirm that artificial intelligence tools were used in the preparation of the submitted work. Specifically, they were used in the following ways:

- improvement of grammar and language style,
- searching for relevant literature and checking references,
- editing and generation of parts of the source code,
- independent quality audit of the textual data extraction pipeline (Section 2.4.4, Table B.1).

Each use of the above methods is documented in Appendix B, which provides a more detailed explanation of which tool was used in which specific part of the work; the author's own contribution is highlighted.

I confirm that:

- my work is original and was prepared in accordance with ethical standards, in particular the Code of Ethics of the Prague University of Economics and Business; that all information sources used are properly cited in the work; and that any generated content has been verified by me;
- I accept full responsibility for the entire content of the thesis.

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Abstract

The risk-factor section (Item 1A) of the 10-K describes the risks a firm faces. This thesis asks whether three properties of the section are associated with stock-return volatility in the weeks after the filing: its length (H1), its share of words from the Loughran–McDonald uncertainty and litigious lists (H2), and the share of its industry peers' vocabulary that the firm does not use (H3). The panel of 5,782 firm-years of 512 large U.S. non-financial firms for fiscal years 2010–2024 is built directly from SEC filings. Regressions of log 30-day post-filing volatility control for pre-filing volatility, size, valuation, past returns and accounting fundamentals, with industry and fiscal-year fixed effects and standard errors clustered by firm and year. Longer Item 1A sections are associated with higher post-filing volatility ($\hat{\beta} = +0.053$, $t = +4.51$) at every horizon from 5 to 365 days, but the association does not hold within firms over time. Holding length fixed, the density of risk words is not significant at 30 days; only the litigious words have a negative association, which is sensitive to the sample. The peer-relative omission gap is not associated with volatility. H1 is supported between firms; H2 and H3 are not supported.

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, risk factors, Item 1A, Loughran–McDonald, LM-Litigious word list, boilerplate disclosure, panel data, corporate disclosures

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List of Abbreviations

API Application Programming Interface
CIK Central Index Key
DV Dependent Variable
EDGAR Electronic Data Gathering, Analysis, and Retrieval
EPS Earnings Per Share
FE Fixed Effects
FY Fiscal Year
FYE Fiscal Year End
GLM Generalized Linear Model
L&M Loughran and McDonald
LLM Large Language Model
MD&A Management's Discussion and Analysis
NCI Non-Controlling Interest
NLP Natural Language Processing
OCF Operating Cash Flow
OLS Ordinary Least Squares
ROA Return on Assets
SEC Securities and Exchange Commission
SIC Standard Industrial Classification
TOC Table of Contents
XBRL eXtensible Business Reporting Language

Introduction

Motivation and context

Annual 10-K reports help investors understand a company's financial position and the risks it faces. Their risk-factor section, Item 1A, provides management's account of potential threats to future performance. These explanations add context that financial figures alone cannot provide. However, more disclosure does not necessarily mean more useful information. A lengthy risk section may describe a complex business, but it may also contain repeated warnings and standard legal language.

Stock-return volatility after a filing provides a way to examine whether these disclosures contain information about subsequent market risk. The question is whether features of the text help explain later volatility beyond what is already observable from a firm's financial characteristics and past volatility.

Research problem and objective

Previous research shows that risk disclosures can contain information about firms' underlying exposures. Campbell et al. (2014) find that firms facing greater risk disclose more risk factors, while Kravet and Muslu (2013) associate increases in risk disclosure with higher subsequent volatility. At the same time, Dyer et al. (2017) document increasing length and repetition in annual reports. Together, these findings motivate separating the amount of disclosure from the language it contains.

This thesis examines whether risk-related wording and differences from industry peers provide additional information beyond disclosure length. Risk-word density captures the proportion of uncertainty and litigious words in Item 1A. The peer-relative omission measure captures vocabulary used by industry peers but absent from the firm's own disclosure. This is an indirect measure of disclosure differences: missing words do not necessarily mean that a company has left relevant risks undisclosed.

The analysis covers large U.S. non-financial firms over fiscal years 2010–2024. It combines text and accounting data from SEC filings with stock-return data and uses panel regressions controlling for pre-filing volatility and other firm characteristics. Comparisons across firms and within firms over time help distinguish persistent differences from changes in disclosure. The analysis identifies associations rather than causal effects.

The main results show that disclosure length has the clearest association with subsequent volatility across firms, while risk-word density and peer-relative omissions provide limited additional evidence. The length association does not hold robustly within firms.

Contributions

The thesis makes a practical and an empirical contribution. First, it develops a reproducible process for extracting risk disclosures and combining them with accounting and market data. Documenting this process makes the dataset construction open to inspection and allows the analysis to be repeated or extended.

Second, the common empirical framework helps assess whether more complex textual measures add information beyond a simple word count. Distinguishing between-firm and within-firm evidence also helps clarify what conclusions can reasonably be drawn from the observed relationships.

Thesis structure

Chapter 1 reviews the relevant literature. Chapter 2 describes the data and their preparation, while Chapter 3 presents the hypotheses and empirical methods. Chapter 4 reports the results, and Chapter 5 examines their robustness. The conclusion discusses the findings, limitations, and directions for further research.

1. Literature Review

Research on corporate disclosures examines whether information provided by management helps investors understand a firm's performance and risk. This chapter focuses on risk-factor disclosures and their relationship with stock-return volatility. It reviews evidence on disclosure length, risk-related vocabulary, and comparisons between firms' disclosures. These studies provide the background for the three hypotheses developed in Chapter 3.

1.1 Risk disclosures and market risk

Risk disclosures can inform investors about uncertain outcomes that are not fully captured by accounting figures. However, providing more information does not necessarily reduce perceived risk. A disclosure may explain an existing uncertainty more clearly, but it may also reveal a risk that investors had previously underestimated. This distinction helps explain why useful disclosure can be associated with higher, rather than lower, subsequent volatility.

Campbell et al. (2014) examine the information content of mandatory risk-factor disclosures in 10-K filings. They find that firms facing greater risk disclose more risk factors and that the types of risks discussed reflect their underlying exposures. The disclosed information is also associated with market-based risk measures, information asymmetry, and firm value. Their findings suggest that risk-factor sections contain information about firms' economic circumstances and cannot simply be dismissed as standard legal text.

Kravet and Muslu (2013) study changes in textual risk disclosures in annual 10-K reports. Increases in disclosure are associated with higher stock-return volatility and trading volume around and after the filing, as well as greater dispersion in analysts' forecast revisions. Their evidence is consistent with disclosures drawing attention to uncertainty rather than necessarily resolving it. Unlike a comparison of disclosure levels across firms, their approach examines annual changes in risk disclosure. These are related but different questions: firms that consistently disclose more risks need not experience higher volatility whenever they expand their disclosures.

Bao and Datta (2014) use a topic model to identify different risk types in 10-K disclosures. They find that many risk categories have no significant relationship with investors' risk perceptions, while informative categories can be associated with either increases or decreases. Their findings suggest that an aggregate disclosure measure may combine information with different implications.

Together, these studies support examining disclosure alongside subsequent market risk, but they do not establish that more disclosure always has the same meaning. They also require care in interpretation. A relationship between disclosure and later volatility may reflect underlying business risk, investors' responses to information, or both. Observing volatility

after a filing does not, on its own, identify the effect of the filing.

1.2 Disclosure length and standard language

Disclosure length is a simple measure of how much a company says about risk. It is relatively easy to calculate and compare across filings. Its main limitation is that it treats all words as contributing equally to disclosure volume, regardless of whether they describe a distinct exposure, repeat earlier statements, or provide general qualifications.

The findings of Campbell et al. (2014) provide a reason to expect a positive relationship between the amount of risk discussion and subsequent risk. However, this does not mean that a word count directly measures the number or severity of a firm's risks. Length can also reflect reporting practices and the way a company organizes its discussion.

Dyer et al. (2017) document changes in 10-K reports between 1996 and 2013. Reports became longer and contained more boilerplate language, repetition, and text retained from earlier years. Their topic analysis shows that disclosures concerning fair value, internal controls, and risk factors account for much of the increase in length. This evidence suggests that growth in disclosure volume can reflect reporting requirements and accumulated text, rather than only changes in economic risk. Their study concerns the full 10-K, so its findings should not be interpreted as a direct test of Item 1A length and subsequent volatility.

Legal considerations offer another explanation for lengthy and standardized disclosures. Cazier et al. (2021) find that longer and more boilerplate risk disclosures are less likely to be judged inadequate in judicial and regulatory reviews. They also find that peer firms are more likely to borrow language after courts have judged it adequate. This shows that shared wording can serve a legal purpose. It does not establish whether that wording provides investors with useful new information.

These studies suggest that length combines several influences. A long section may reflect greater exposure to risk, more extensive explanation, or a cautious reporting style. H1 therefore tests whether disclosure volume is associated with subsequent volatility, without treating length as a direct measure of disclosure quality. Examining both differences between firms and changes within firms helps assess whether the association reflects persistent company characteristics or changes in disclosure over time.

1.3 Risk-word density and disclosure content

Textual analysis offers ways to examine what companies write, rather than only how much they write. A common approach counts words from predefined dictionaries. The choice of dictionary matters because words can have different meanings in financial reporting and everyday language.

Loughran and McDonald (2011) show that general-purpose dictionaries can misclassify common financial terms. They develop word lists suited to financial text, including uncertainty and litigious categories. The uncertainty list captures imprecision and uncertainty, while the litigious list includes legal language that does not always indicate an actual dispute. In their full-report analysis, uncertainty-word proportions are positively associated with subsequent volatility, while the result for litigious wording depends on how words are weighted. This provides a reason to examine the two categories separately rather than assume that they carry identical information.

A dictionary measure remains an approximation. Counting selected words does not fully capture the meaning of the sentences in which they appear. The same term may occur in a general warning or a detailed explanation of a specific event. Loughran and McDonald (2016) emphasize the measurement and implementation challenges involved in textual analysis. Their discussion supports interpreting textual variables according to what they directly measure, rather than assigning them broader meanings without validation.

This distinction is particularly important for disclosure specificity. Hope et al. (2016) study how precisely firms describe disclosed risks. They find that more specific risk-factor disclosures are associated with stronger market reactions to 10-K filings and that analysts are better able to assess fundamental risk. Their findings support the usefulness of detailed disclosure, but they do not establish that a higher proportion of uncertainty or litigious words measures greater specificity. Nor do they imply that useful risk disclosure must be associated with lower subsequent volatility.

For the present thesis, risk-word density measures the share of Item 1A vocabulary belonging to the selected dictionaries. It is not a direct measure of clarity, specificity, or completeness. A higher density may indicate more discussion of uncertain outcomes and legal exposures, but it may also reflect standard cautionary wording. The latter possibility could weaken the relationship with volatility; it does not by itself predict a negative relationship.

Length and density must also be considered together. A section can contain more risk-related words simply because it contains more words overall. Dividing by section length changes the question from the amount of risk wording to its share of the text. H2 examines whether this composition provides additional information after controlling for length. Given the differences between prior measures and the present specification, the hypothesis does not impose a direction on the association.

1.4 Peer comparisons and disclosure omissions

A disclosure can also be evaluated relative to what other firms report. Industry peers provide a possible benchmark because they may face similar business conditions and exposures. Such a comparison asks whether a firm covers vocabulary commonly used by comparable companies, rather than whether its own disclosure is long or contains many dictionary words.

Peer comparisons already appear in the risk-disclosure literature. Kravet and Muslu (2013) distinguish disclosure patterns shared within an industry and year from those that differ at the firm level. Their results are less pronounced for disclosures that deviate from the industry-year pattern. This indicates that the relationship between a firm's language and its peers' language can matter when interpreting disclosure measures. It does not, however, establish that missing peer vocabulary represents withheld risk information.

Selecting suitable peers is itself a measurement issue. Hoberg and Phillips (2016) use product descriptions in 10-K filings to construct firm-specific competitor groups that change over time. Their approach illustrates how textual information can identify relationships that broad industry classifications may miss. Their study concerns product markets, rather than omissions from risk-factor sections, but it highlights a relevant limitation: firms assigned to the same industry need not have identical businesses or risk exposures.

Comparisons of text can also be affected by document length. Brown and Tucker (2011), studying annual changes in MD&A, account for the mechanical relationship between textual similarity and length. They find that disclosure revisions are related to changes in firms' economic circumstances and to filing-date price responses. Although their comparison is across years within a firm, rather than across peers, it provides a useful methodological lesson: a text-difference measure may partly reflect the amount of text available for comparison.

The omission measure in this thesis extends these ideas by examining the weighted share of peer vocabulary absent from a firm's Item 1A. One possible interpretation is that a larger gap reflects incomplete discussion of relevant exposures. If so, it may be associated with greater uncertainty and higher subsequent volatility. This is the proposed reasoning behind H3, rather than an established finding of the studies reviewed above.

Several alternative explanations remain possible. A firm may not face the same risks as its peers, may describe them using different words, or may have a shorter section. Shared peer vocabulary may also reflect imitation rather than shared exposure. The measure therefore captures a vocabulary gap, not verified omissions of economic risks or deliberate silence. Testing its relationship with volatility requires accounting for disclosure length and examining whether the result is sensitive to how the gap is measured.

1.5 Synthesis and link to the hypotheses

The literature supports studying risk disclosures as a source of information about firm risk, while showing that different textual features capture different aspects of reporting. Disclosure volume can reflect underlying exposure, but also reporting practices. Dictionary measures describe vocabulary composition, while specificity concerns the detail with which risks are explained. Peer comparisons provide another perspective, although differences in wording do not necessarily indicate missing information.

This thesis brings three measures into a common regression framework. H1 examines the association between Item 1A length and subsequent volatility. H2 asks whether risk-word density adds information after length is accounted for, including whether uncertainty and litigious wording behave differently. H3 examines whether a peer-relative vocabulary gap provides additional information beyond both measures.

The study is therefore a focused comparison and extension of existing approaches. It does not assume that any single measure captures overall disclosure quality. Chapter 3 translates these questions into variable definitions, formal hypotheses, and empirical specifications.

2. Data

This chapter describes how the firm-year panel is built. Financial variables come from the SEC EDGAR Company Facts API, the narrative sections are parsed from the HTML of the filings on EDGAR, and return-based measures are computed from daily prices downloaded from Yahoo Finance with the `yfinance` package. No commercial research database (such as Compustat, CRSP or a vendor 10-K corpus) is used. Building the data from these sources means that the textual and financial data can be matched to the same filing, period and reporting entity, and that the processing steps are visible in the code that accompanies the thesis. It also means that problems that commercial providers usually handle (XBRL tag heterogeneity, irregular HTML, amendments, share counts of multi-class firms) are dealt with directly; this chapter describes those choices and their consequences. Yahoo Finance prices are not raw: the adjusted close used for returns is adjusted for splits and dividends by the provider, and the price series were downloaded once and kept fixed for all computations. The definition, source and timing of every variable are listed in Appendix A.

Figure 2.1 gives an overview. The pipeline starts from the SEC sources, applies the sample filters, and runs a financial (XBRL) branch and a textual branch that are merged into the firm-year panel.

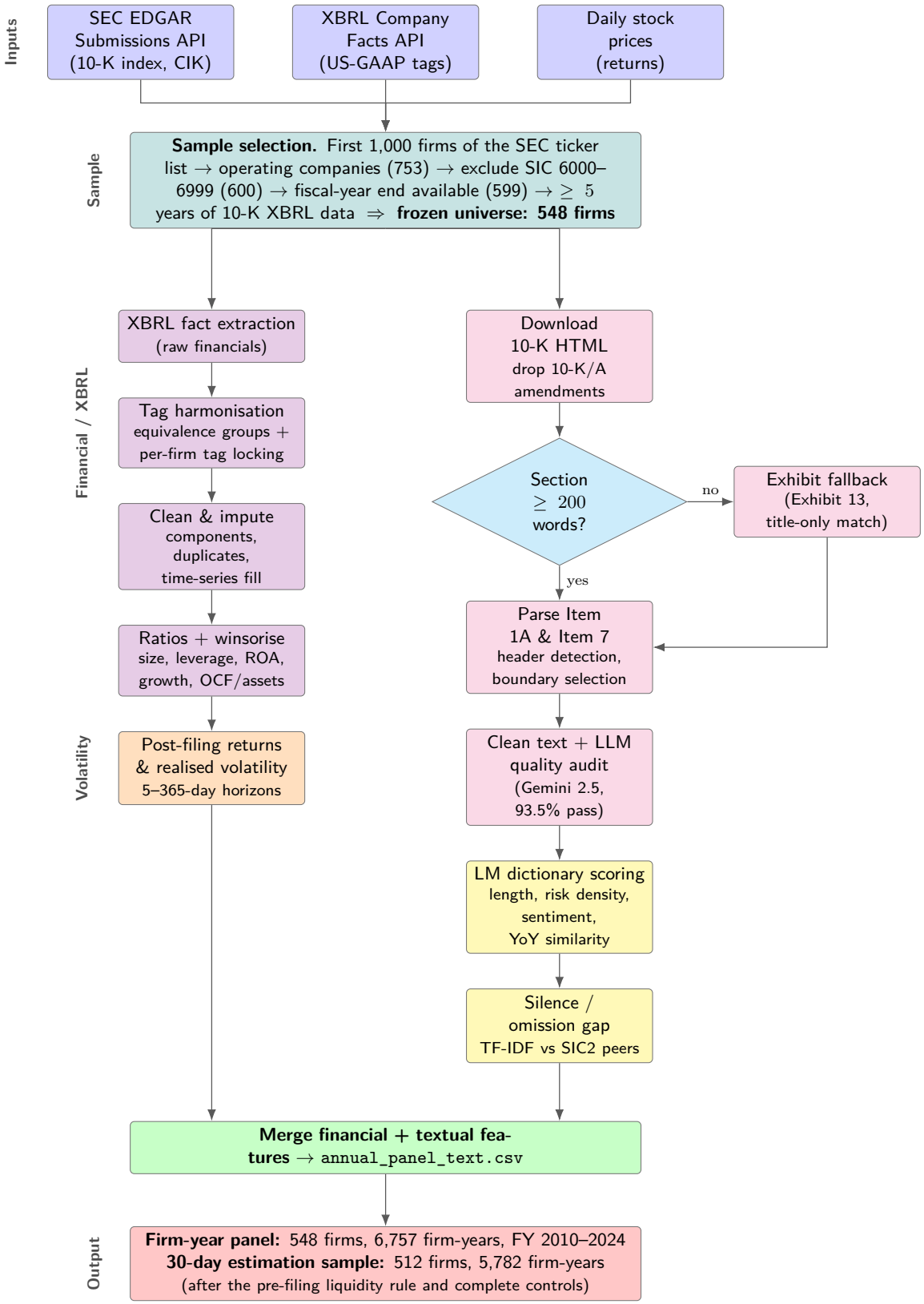


Figure 2.1: Operational pipeline for the complete data preparation. The workflow runs top-to-bottom from the raw SEC sources through sample selection, then splits into a financial (XBRL) branch and a textual (10-K narrative) branch that are merged into the final regression-ready firm-year panel. Own processing.

2.1 Sample selection

The starting universe is the first 1,000 entries in the SEC file `company_tickers.json`, which maps stock tickers to Central Index Keys (CIKs). The version used in this study was archived in the project repository in October 2026. Four filters are then applied (Table A.1 in Appendix A). First, only entities classified by the SEC as operating companies are retained, which excludes exchange-traded funds, closed-end funds, and shell companies (1,000 to 753). Second, firms with SIC codes 6000–6999 (finance, insurance, and real estate) are excluded. Following Fama and French (1992), financial firms are omitted because leverage and related balance-sheet ratios have a different economic interpretation for such firms than for non-financial operating companies, as financial assets and liabilities constitute a core component of their business model rather than primarily a financing choice. This filter reduces the sample from 753 to 600 firms. Third, firms without a fiscal-year end date in the metadata are dropped (600 to 599). Fourth, firms must have at least five fiscal years of available 10-K XBRL data. This requirement ensures that lagged variables, such as asset growth, can be constructed while still leaving at least four years of usable panel observations. The filter reduces the sample to 550 firms using current metadata; the frozen universe used in the final analysis contains 548 firms. These firms form the panel of 6,757 firm-years for fiscal years 2010–2024, and the 30-day estimation sample is the 5,782 firm-years of 512 firms that remain after the requirements below (Table A.1). Figure 2.1 shows the same stages.

To avoid conditioning sample inclusion on outcome-period information, filings are not filtered on the basis of post-filing return availability. An earlier version of the data pipeline excluded firms with valid post-filing volatility for fewer than 90% of their filings. Because this criterion relied on price data from the outcome period, it could induce selection bias and was therefore removed. Instead, a filing enters the regression sample only if at least 200 daily closing prices are available during the 252 trading days preceding the filing date. This requirement excludes thinly traded and recently listed firms using only information available at the time of filing and affects 1.3% of filings.

Sample size and look-ahead. Large firms have complete machine-readable filing histories, are liquid, and their filings are consistently formatted, which keeps measurement error in both branches low. The cost is that the list is drawn from today's large firms: firms that failed, were acquired or shrank during 2010–2024 are missing, and the panel grows from 152 firm-years in fiscal year 2010 to over 500 from 2020. The results therefore describe large surviving U.S. firms (Section 3.6). The 30-day estimation sample has 5,782 firm-years of 512 firms. By comparison, Campbell et al. (2014) study risk-factor disclosures of a broad cross-section of U.S. firms and Loughran and McDonald (2011) use about 50,000 10-K filings; their aims require broad coverage, whereas the within-industry-year comparisons here rely on a balanced set of large issuers.

2.2 Annual filings as the unit of observation

We use annual 10-K filings rather than quarterly 10-Q reports. A quarterly panel would have roughly four times as many observations, but two problems appeared in small pilot tests (Tables 2.1 and 2.2); the pilots are illustrations on a handful of firms, not representative estimates.

Financial data in the fourth quarter. 10-Q filings cover only the first three quarters, so a quarterly panel must reconstruct the fourth quarter as the annual figure minus the three quarters. In a pilot of 30 non-financial firms (Table 2.1), all three quarterly operating-cash-flow facts were available in only 7% of firm-years, because most firms report cumulative year-to-date cash flows. Operating cash flow enters the control vector, so a quarterly panel would lose it at the source. For net income and operating income the implied and the explicitly reported fourth quarter differ by more than 1% of the annual figure in about 10 to 14% of comparable firm-years. The residual absorbs year-end audit adjustments that are booked in the fourth quarter, whereas a later filing may report a recast figure, so the two represent different versions of the same accounting history.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: "Q1 – Q3" indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. "Explicit Q4" indicates the share with a standalone fourth-quarter fact (period ≈90 days ending at fiscal year-end). "Comparable" is the count of firm-years for which both an explicit Q4 fact and the implied residual are available. "Match" reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Risk factors in quarterly reports. Form 10-Q (Part II, Item 1A) requires only *material changes* to the risk factors disclosed in the latest 10-K, whose content is governed by Item 105 of Regulation S-K (17 CFR 229.105). In a pilot of the six most recent 10-Q filings of 15 large firms, 52% contained a substantive risk-factor section and the rest referred back to the annual report (Table 2.2; Table A.7 lists the firms). Text variables measured on the full risk-factor section therefore cannot be built quarterly.

Annual filings contain audited full-year accounting data and complete risk-factor and MD&A sections, and they are the unit of analysis in the 10-K textual-analysis literature (Li, 2008; Loughran & McDonald, 2011). Matching the observation interval to the reporting frequency of the documents also means that changes in the text variables correspond to actual revisions of the annual disclosure.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
MD&A (Management's Discussion and Analysis)		
Section found (%)	93	100
Median word count	20,494	5,806
Risk Factors		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: "Found" requires >200 words extracted by regex-based section parser. "Substantive" for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., "no material changes from those disclosed in our Annual Report"). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

2.3 Financial data from XBRL

Point-in-time values. For each firm and fiscal period the panel takes the values, the filing date and the accession number of the *first* 10-K that reports facts for that period end, filed within 180 days of it. Restated figures published later (in a 10-K/A or as comparatives in the next year's 10-K) are not mixed in, because they were not public when the post-filing window starts. A 10-K/A is used only when no original 10-K reports the period. All variables of a firm-year therefore come from one filing, identified by its accession number, which also identifies the filing whose text is used. The fiscal year is the calendar year in which the period ends (periods ending on 1–7 January count to the previous year, for firms with 52/53-week years).

Fiscal years and calendar time. Firms do not share a common fiscal calendar, so 10-K filings from the same fiscal year are released at different points in calendar time (Table A.3 in Appendix A). In the estimation sample, 72.6% of firm-years end in December, while 145 of the 512 firms have fiscal year-ends between January and November. The median lag between fiscal-period end and filing is 53 days (10th to 90th percentile: 38 to 59 days), which places most filings in February. This concentration reflects the sample's focus on large firms, which typically file annual reports relatively soon after fiscal year-end. Despite this clustering, filing dates for a given fiscal year are dispersed over more than a full calendar year. To ensure comparability across firms, all timing variables are anchored to each firm's own filing date. Post-filing volatility windows, pre-filing volatility windows, and the lagged-volatility control are all defined relative to the firm's filing day rather than a common calendar date (Section 3.1). Accounting variables and textual measures are likewise taken from the filing

that opens the measurement window. Fiscal years are assigned using the actual reporting-period end rather than the fiscal-year-end month currently reported by the SEC, which proved unreliable for some 52/53-week fiscal-year firms and had shifted the fiscal-year labels of approximately 20 firms in an earlier version of the panel. In addition, lagged observations are matched only when the previous filing corresponds to the same fiscal year one year earlier, preventing changes in fiscal-year ends from creating lags of unequal length. Finally, because fiscal-year fixed effects cannot fully absorb calendar-time shocks when filings from the same fiscal year are spread across many months, the robustness checks replace fiscal-year fixed effects with filing-month fixed effects and restrict the sample to December year-end firms (Section 5.2).

Tag choices. One accounting concept can be reported under several US-GAAP elements, so the elements must be mapped explicitly (Chychyla & Kogan, 2015). Two kinds of alternatives are handled differently.

- *Equivalent tags* measure the same quantity and are used interchangeably. Net income is measured on a consolidated basis: `ProfitLoss` (including non-controlling interests, NCI) where a firm reports it, and `NetIncomeLoss` otherwise, which under US GAAP is then the same consolidated figure. This makes ROA consistent with total assets, which also include NCI.
- *Approximately equivalent tags.* The two operating-cash-flow tags (total and continuing operations) differ when a firm has cash flows from discontinued operations, so they are not strictly equivalent. The panel uses the total-operations tag where it exists and the continuing-operations tag otherwise, without locking, and accepts the difference as measurement error in one control variable. The size of this error is reported below.
- *Non-equivalent tags*, such as stockholders' equity with or without NCI, or income from continuing operations versus net income, are *locked* per firm: each firm keeps the tag it reports in most years, and values under other tags are set to missing. The choice uses the firm's whole history, a mild look-ahead in which tag is used, not in the values.

Table A.4 in Appendix A reports the tag choices, the values lost to locking (637 of 7,900 equity values, 38 net-income values, no cash-flow values) and the differences between paired tags in firm-years that report both. Where both net-income tags are present (3,838 firm-years), the median absolute difference is 0.043% of total assets, but the 99th percentile is 3.56% and the maximum 37.4%; in 303 firm-years (7.9%) the difference exceeds 1% of total assets, which we call material. These differences are the NCI share of income, which the consolidated definition now includes consistently. For the two operating-cash-flow tags (417 firm-years) the difference exceeds 1% of assets in 50 firm-years (12.0%), with a median of 0.035%. The cash-flow differences are therefore material in a non-trivial share of the overlapping firm-years, which is why the two tags are not called equivalent above. Only 18 firms lock on the continuing-operations tag, and operating cash flow over assets is one of nine controls.

Missing balance-sheet values. If total liabilities are missing, they are computed as total assets minus stockholders' equity (2,171 firm-years; 1,935 in the 30-day regression sample). Under ASC 810 (formerly SFAS 160) non-controlling interests are part of equity, so the identity holds only if the equity figure includes NCI. This is the case for the 315 firms whose locked equity tag is `StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest`. For the 228 firms whose locked tag is the parent-only `StockholdersEquity`, the computed liabilities also contain any NCI. This affects 626 of the 1,935 imputed firm-years in the regression sample (60 firms). The data do not show how many of these firms have non-zero NCI. Where they do, leverage is overstated by the NCI share of total assets. The check without imputed values in Section 5.2 removes all of these firm-years. If total assets or liabilities are still missing, the value of the immediately preceding fiscal year is carried forward (1 and 32 firm-years); values are never filled from later years. Each imputation is flagged, and Section 5.2 re-estimates the models without imputed values.

Book equity. Book-to-market uses book equity attributable to the parent (`StockholdersEquity`), which matches the market capitalisation of the parent's common shares. When this tag is missing, total equity including NCI is used (430 of 5,782 firm-years in the 30-day sample), and when that is missing too, total assets minus total liabilities (55 firm-years). The other 5,297 firm-years use parent equity.

Market capitalisation. Shares outstanding are taken from the same 10-K as the other variables, matched by accession number: the cover-page count (`dei:EntityCommonStockSharesOutstanding`, summed over share classes), else the balance-sheet count, else the fiscal-year weighted-average basic count. The last source is needed for multi-class firms whose cover-page counts are reported per class and are not available in the Company Facts API (445 firm-years); a robustness check drops these firm-years. Shares are multiplied by the as-traded closing price on the last trading day before the filing, which is rebuilt from the split-adjusted price and the split history.

2.4 Text extraction from 10-K filings

The study uses Item 1A (Risk Factors) and Item 7 (Management's Discussion and Analysis, MD&A) of the 10-K.

2.4.1 Source of the text

For each original 10-K in the panel the primary HTML document was downloaded from the EDGAR archive, using the document name listed in the EDGAR Submissions API.[1] Where

[1] `https://data.sec.gov/submissions/CIK{cik}.json`; see https://www.sec.gov/search-filings/edgar-application-programming-interfaces.

the filing incorporates a section by reference to an exhibit, the exhibit was fetched from the same filing (Section 2.4.2). All requests carried the declared user-agent header that the SEC requires and stayed below its limit of ten requests per second. The extraction was run once, in October 2026; the stored texts were not re-extracted for this version, and each stored section is linked to the accession number of the filing it came from. In seven firm-years the stored text comes from a 10-K/A whereas the panel now uses the original 10-K; these firm-years are excluded from all text regressions.

2.4.2 Section identification and exhibit fallback

The text of a 10-K is embedded in HTML without standard tagging, mixed with table-of-contents entries, running headers and cross-references, so a plain search for "Item 1A" typically finds the table of contents. The parser works on the document structure before the text is flattened:

1. **Candidate detection.** Elements (paragraphs, divisions, table cells, headings) whose text matches an "Item X" pattern are kept as candidate headers if they have at most 25 words, carry a heading cue (bold font, all capitals, or a font size of at least 12 pt), and contain no hyperlink (table-of-contents links).
2. **In-place tagging.** Each candidate is marked inside the HTML tree, and only then is the document converted to text, so header positions survive the conversion.
3. **Candidate selection.** A candidate for Item 1A must have the word "risk" within 300 characters, and one for Item 7 "management", "discussion" or "MD&A". Candidates with at least three other item headers within 300 characters (a table of contents) and candidates preceded by "see", "refer to" or similar phrases are discarded when alternatives exist. Among the rest the parser takes the candidate followed by the longest section.
4. **Section bounding.** The section ends at the next item header: Item 1B, 1C or 2 for Item 1A, and Item 7A, 8, 9, 9A or 9B for Item 7. Several terminators are needed because some items are optional (Item 1B applies only to some filers, and Item 1C exists only for fiscal years ending on or after 15 December 2023). A section needs at least 200 words.

Figure 2.2 illustrates the four steps on a simplified DOM tree.

The parser is intentionally conservative: it does not operate on a flattened HTML string, but on the DOM tree, where heading structure and placement can be tracked before the text is reduced to plain running text. The same logic is also used for the exhibit fallback described below. The implementation details of the candidate-detection and fallback logic are reported in Appendix A.4.

Exhibit fallback. If no section of at least 200 words is found and the text under the header (fewer than 500 words) refers to an exhibit or the annual report, the parser applies the same

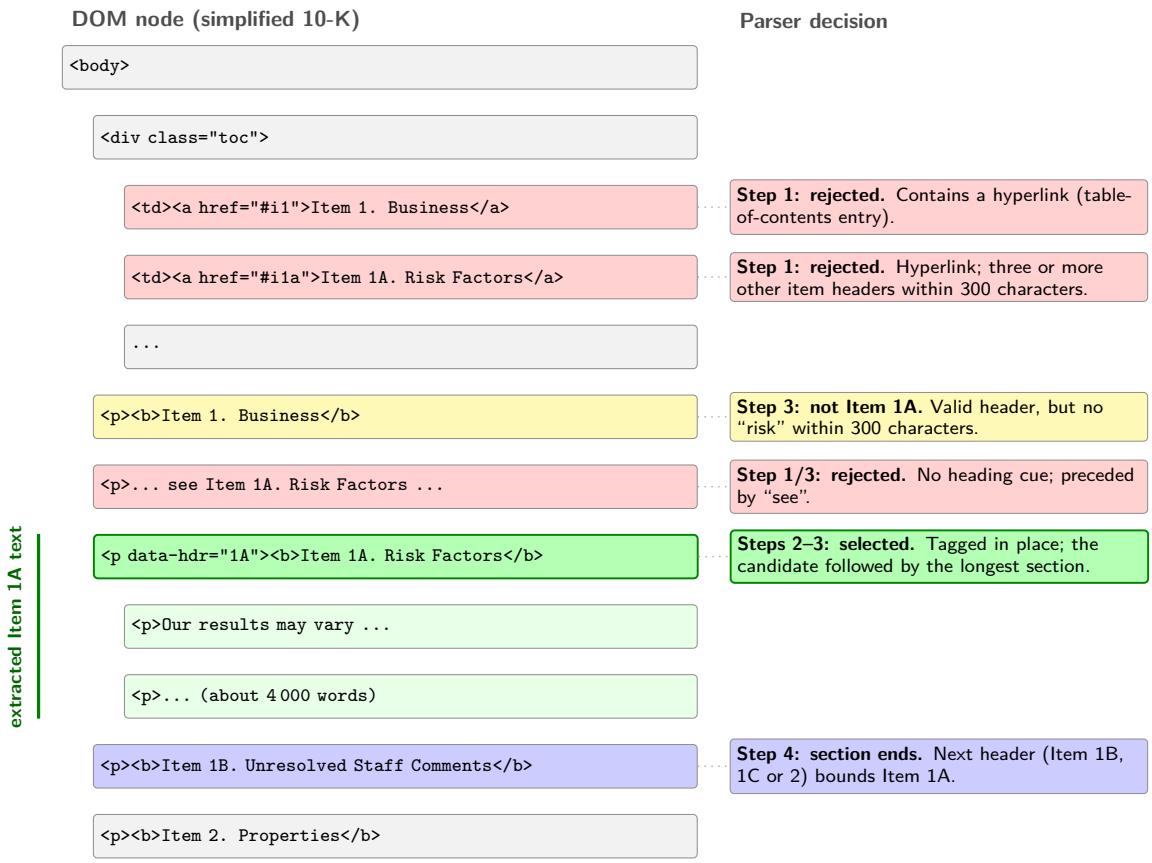


Figure 2.2: How the parser reads a simplified 10-K DOM tree for Item 1A. Table-of-contents entries and cross-references are rejected, the real header is tagged in place and selected, and the next item header ends the section. Step numbers refer to the list above. Own processing.

procedure to the referenced exhibit (usually Exhibit 13), using title patterns ("Risk Factors", "Management's Discussion and Analysis") because exhibits have no item numbers. It does not read the page references given in the filing. The extraction run did not record which sections came from an exhibit, so the share cannot be measured without downloading the filings again.

2.4.3 Post-processing and tokenisation

During the HTML conversion, scripts, style sheets and the hidden inline-XBRL header are removed, and tables are removed if they are numerical: at least 30% of their non-empty cells (with at least four cells and two rows) are bare numbers or currency amounts, or more than 40% of their characters are digits or number symbols. Other tables, which in many filings are used for page layout, keep their text. After the section has been bounded, the parser removes running "Table of Contents" headers, page footers such as "2019 Form 10-K 59", standalone page numbers, residual numeric table rows (lines of at least four tokens of which more than 40% are numbers), and short lines that repeat at least four times (running headers), and replaces non-breaking spaces. Footnote markers and text of layout tables are not removed.

For the dictionary measures the text is lowercased and split into tokens of at least two letters (digits, hyphens and apostrophes split tokens); a section with fewer than 100 tokens is treated as missing. The category definitions and financial-text motivation follow Loughran and McDonald (2011); the word-level input is the Loughran–McDonald Master Dictionary, 1993–2025 release (Loughran & McDonald, 2025). I count words with a positive year-added category code and exclude words marked with a negative removal code. The resulting lists contain 297 uncertainty and 903 litigious words; the two lists share no word.

2.4.4 Coverage

Among the 6,757 firm-years in the panel, 96.3% have an Item 1A section with at least 100 tokens, and 94.9% have an Item 7 section of the same minimum length (Table A.2 in Appendix A). Coverage is lowest in fiscal year 2010 (92.8%), and it rises to 94.7% in 2024. The lower 2024 figure reflects a correction to the fiscal-year labels of about 15 firms with 52/53-week calendars; their fiscal-year-2024 filings had not yet been extracted.

Extraction audits. Two large language models (LLMs) independently audited a stratified random sample of stored sections. The sample contains five three-year strata (2010–2012 to 2022–2024), with 20 sections of each type per stratum, for 200 sections in total. Item 1A and Item 7 were drawn independently, so the 200 sections come from 197 filings. Each section was rated on five criteria: correct section, start boundary, end boundary, content quality and completeness. Ratings were recorded as Pass, Minor issue or Fail. The first audit, by Google Gemini 2.5 Flash, saw the full extracted section; the second, by Claude Opus 5.5, saw the

first and last 1,200 characters, a middle excerpt and the word count, without the first audit's ratings. Neither model saw the rest of the filing, so the boundary judgments were based only on the extracted text. Gilardi et al. (Gilardi et al., 2023) find that ChatGPT can outperform crowd workers on several tweet-annotation tasks. This motivates exploring LLM assistance, but does not validate extraction boundaries or completeness in 10-Ks. Their judgments are therefore not a substitute for a human reading of the filings. I also checked 51 sections by hand against the original EDGAR filings, without seeing the LLM ratings: 50 were drawn at random within the same strata, and one additional section was flagged by an LLM.

Under the first audit, 187 of 200 sections (93.5%, Wilson 95% interval 89.2–96.2%) were acceptable on all five criteria (Pass or Minor), and 179 (89.5%) were rated Pass on all five. The 13 sections with a Fail include five with no extracted text. By stratum, five fall in 2010–2012, three in 2013–2015, two each in 2016–2018 and 2019–2021, and one in 2022–2024. Under the second audit, 190 sections (95.0%, 91.0–97.3%) were acceptable. Both audits reject the same eight sections (Table B.1 in Appendix B). In the manual check, all 51 sections were the correct section. Four had a boundary error: two started inside Item 1, one ran a few lines into the financial statements, and one ran through Item 8. Three of those errors were among the 50 random sections, implying an error rate of 6.0% (Wilson 95% interval 2.1–16.2%). The manual verdict on the three criteria matches the first audit on all 51 sections and the second on 50. Errors are more common in early filings, whose structure is less standardised.

Relation to other approaches. The content and volume of 10-K disclosures have changed over time (Dyer et al., 2017), while parsing procedures are not always documented in sufficient detail to facilitate replication (Loughran & McDonald, 2016). Three design choices are therefore set out clearly here.

First, Loughran and McDonald (2011) analyse both filing-level text and the MD&A subsection. Section-level extraction requires identifying item boundaries, which is difficult because of table-of-contents entries, cross-references and inconsistent headings (Loughran & McDonald, 2016, Section 6.4). The DOM-based parser used here keeps the HTML structure intact before converting the document to text and filters candidate headings using formatting cues, hyperlinks, local context and keywords. Its limitation is that headings without the expected formatting characteristics may not be identified.

Second, mandatory XBRL reporting is not a complete substitute for extracting narrative disclosures. The SEC's 2009 XBRL amendments covered financial statements but did not require interactive data for MD&A or other narrative disclosures (U.S. Securities and Exchange Commission, 2009, Release No. 33-9002). The present study therefore extracts MD&A and Risk Factors directly from original EDGAR filings across the 2010–2024 sample period.

Third, restricting extraction to the primary Form 10-K document can omit disclosures incorporated by reference from annual reports to security holders. General Instruction G(2) permits information required by Parts I and II, including Item 7 (MD&A), to be incorporated by reference. Note 2 requires the incorporated material to be filed electronically as an exhibit

(U.S. Securities and Exchange Commission, 2014, General Instruction G(2), Note 2); see also 17 C.F.R. § 232.303(b) (United States, 2026a) and § 240.12b-23 (United States, 2026b). This practice is illustrated by the 10-K filings of Sonesta International Hotels Corporation (2010), First Community Financial Corporation (2012) and Wal-Mart Stores, Inc. (2014), all of which incorporate MD&A from Exhibit 13 (First Community Financial Corporation, 2012; Sonesta International Hotels Corporation, 2010; Wal-Mart Stores, Inc., 2014). Loughran and McDonald (2011) exclude incorporated MD&A from their section-level analysis because exhibit boundaries are difficult to identify. The exhibit-fallback procedure used here instead attempts to recover referenced disclosures when the primary section is missing or too short.

The practice was uncommon in the 2014 cross-section: roughly 2% of registrants incorporated some Part I or Part II information from their annual reports, with Part II incorporation more common than Part I (U.S. Securities and Exchange Commission, 2016, p. 300, footnote 1007). This figure does not measure MD&A incorporation specifically or establish historical trends.

Overall, the parser uses document structure to reduce section-identification errors while retaining the risk that headings with atypical formatting may be missed.

3. Methodology

This chapter describes the regressions I use to test the three hypotheses. In every test, I regress post-filing realised volatility on features of the Item 1A risk-factor section, a common set of controls that are known when the 10-K is filed, and fixed effects. Section 3.1 defines the dependent variable and explains when each variable is measured. Section 3.2 lists the controls. They are the same in every specification, so the text coefficients can be compared across hypotheses. Sections 3.3 and 3.4 cover the joint test of the volume channel (H1) and the composition channel (H2). I estimate them together because the density measure for H2 has section length in its denominator. Section 3.5 introduces the peer-relative omission gap for H3 and the decision rule I use to judge it. Section 3.6 covers estimation and inference.

3.1 Outcome variable and timing

The dependent variable is the natural logarithm of annualised post-filing stock-return volatility measured over a window of h calendar days:

$$\ln \sigma^{[h]}_{i,t+1} \;=\; \ln\!\left(\sqrt{252}\cdot \mathrm{sd}\big(r_{i,d}\big)_{d\in W^{[h]}_{i,t}}\right), \tag{3.1}$$

where $r_{i,d} = \ln(P_{i,d}/P_{i,d-1})$ is the daily log return of firm i computed from dividend- and split-adjusted closing prices, and $W^{[h]}_{i,t}$ is the post-filing window that belongs to the 10-K for fiscal year t. The subscript $t+1$ only means that the window comes after the filing; it does not refer to the next fiscal year. Day 0 is the first trading day on or after the filing date. The window starts two trading days after day 0, so the market reaction on the filing day and the day after is left out, and it then runs for h calendar days. Only returns between two prices inside the window are used. A window needs a minimum number of daily returns (3 for $h = 5$, 5 for $h = 10$, 14 for $h = 30$, and 44, 89 and 199 for the longer windows). If there are fewer, or if the standard deviation is zero, the volatility is treated as missing. A window that would end after the last trading day in the price data (30 October 2026) is also treated as missing instead of being cut short. With this cut-off, every fiscal-year 2024 filing has a complete 365-day window (Section 2.1).

Figure 3.1 shows the timing. Every control and every text variable uses only information that is public at day 0. The accounting variables and the text come from the same 10-K, and the price-based controls use windows that end on the last trading day before the filing date.

A coefficient β on a regressor x means that volatility changes by $100(e^{\beta\Delta x}-1)\%$ when x changes by Δx. The main horizon is $h = 30$ calendar days. I chose it to keep the outcome close to the filing, so that later news has less time to affect it. Earlier studies that link 10-K text to later stock-return volatility have used windows of about one year (Campbell et al., 2014; Kogan et al., 2009; Loughran & McDonald, 2011). I report the 365-day window alongside the main result and use $h \in \{5, 10, 90, 180\}$ as robustness checks.

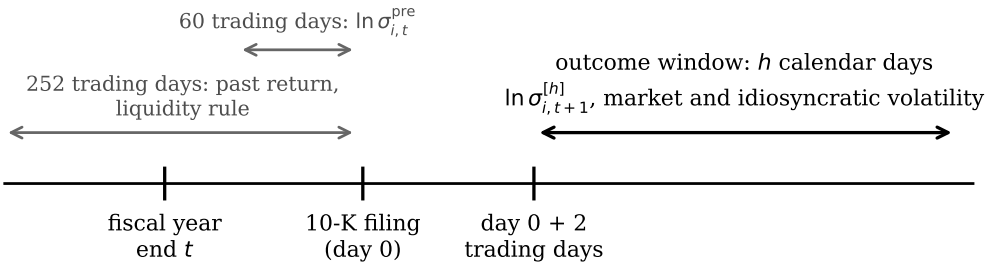


Figure 3.1: Timing of the variables for the 10-K of fiscal year t (not to scale). The outcome window starts two trading days after day 0 and lasts h calendar days; pre-filing volatility uses the 60 trading days, and the past return and the liquidity rule the 252 trading days, that end on the last trading day before the filing date.

3.2 Control variables

Every regression for H1, H2 and H3 includes the same vector $\mathbf{X}_{i,t}$ of nine controls. All of them are known at day 0:

- pre-filing volatility $\ln \sigma^{\text{pre}}_{i,t}$: the log of the annualised standard deviation of the 60 daily returns ending on the last trading day before the filing date;
- firm size ln(Market cap): the as-traded closing price on the last trading day before filing times the shares outstanding reported in the same 10-K;
- book-to-market: book equity attributable to the parent over market capitalisation (total equity including NCI where parent equity is not reported; Section 2.3);
- the past 12-month log return, over the 252 trading days before filing;
- leverage (total liabilities over total assets), return on assets (consolidated net income over total assets), asset growth, the current ratio, and operating cash flow over total assets.

Pre-filing volatility is the most important control. Stock-return volatility is persistent (Kogan et al., 2009), and the way a firm writes its disclosure may reflect conditions that were already visible before the filing. Firm size, book-to-market, past returns and leverage also appear in earlier risk-disclosure regressions (Campbell et al., 2014), although the variables are defined differently. The remaining accounting ratios control for profitability, growth, liquidity and operating cash flow. Operating cash flow over assets is part of the main specifications and not only a robustness check.

An earlier version of this model used a different dynamic control: the volatility of a window of the same length after the previous year's 10-K. Once pre-filing volatility is included, this lag adds nothing. Its coefficient is +0.031 with $t = +0.69$ in the industry fixed-effects model (Table 5.4 reports the firm fixed-effects version), so I report it only as a robustness check.

Each firm's first fiscal year is not in the sample, because asset growth needs the previous year's total assets. Financial ratios, book-to-market and the past return are winsorised at the 1st and 99th percentiles of each regression sample. Market capitalisation and pre-filing volatility enter in logs and are not winsorised.

3.3 H1 — Item 1A length and disclosure volume

Following Campbell et al. (2014), I measure disclosure volume as the logarithm of the Item 1A word count. Campbell et al. find that firms with longer risk-factor sections have higher market-based risk measures later on. This fits the idea that the amount of risk discussion carries information about the firm's risk. The volume variable is

$$\text{Words}_{i,t}^{1A} \;=\; \text{number of tokens in Item 1A of firm } i\text{'s fiscal-year-}t \text{ 10-K}, \tag{3.2}$$

where a token is a run of at least two letters in the lowercased text (Section 2.4.3). The cleaning removes running page headers, page numbers and numeric table rows. It keeps the text of layout tables and footnote markers, so the count is close to the narrative content but not identical to it. The median Item 1A in the estimation sample has 8,700 tokens (Table 4.1).

H1. *If the composition of Item 1A and the controls are held fixed, a longer Item 1A is associated with higher post-filing volatility*: $\beta^{\text{len}} > 0$ in

$$\begin{aligned}\ln \sigma_{i,t+1}^{[h]} \;=\; \alpha \;+\; \boldsymbol{\gamma}'\mathbf{X}_{i,t} \;+\; \beta^{\text{len}} \ln(\text{Words}_{i,t}^{1A}) \\ +\; \beta^{\text{dens}}\,\text{RiskDensity}_{i,t} \;+\; \mu_{j(i,t)} \;+\; \tau_t \;+\; \varepsilon_{i,t},\end{aligned} \tag{3.3}$$

where μ_j is a fixed effect for the two-digit SIC industry of the firm at the filing date, τ_t is a fiscal-year fixed effect, and RiskDensity is defined in Section 3.4. Length and density enter together because length is in the denominator of the density.

3.4 H2 — Risk-word density

H2 is about the composition of Item 1A at a given length. It uses the share of words from the uncertainty and litigious dictionaries of Loughran and McDonald (2011), which were built for financial reporting:

$$\text{RiskDensity}_{i,t} \;=\; \frac{n_{i,t}^{\text{unc}} + n_{i,t}^{\text{lit}}}{\text{Words}_{i,t}^{1A}}, \tag{3.4}$$

where n^{unc} and n^{lit} count the tokens of Item 1A that belong to each list. The two lists share no word in the dictionary version I use (Section 2.4.3), so no token is counted twice.

Previous studies do not give a clear prediction for how this density measure relates to volatility, so H2 is two-sided. I consider two possible explanations.

Risk content. Uncertainty and litigious words may describe uncertain outcomes and legal risks. Campbell et al. (2014) find that risk-factor disclosures reflect the risks firms face, and Kravet and Muslu (2013) find that increases in risk disclosure go together with increases in stock-return volatility. If a higher density of risk words means the section says more about risk, it should be associated with higher volatility.

Standard language. Risk disclosures may also contain general language that firms repeat or copy. Cazier et al. (2021) show that firms have reasons to use standard risk language and sometimes borrow it from other firms. Dyer et al. (2017) document repetition and boilerplate in 10-K reports, and Hope et al. (2016) find that more specific risk disclosures are useful to investors and analysts. A high share of risk words may therefore reflect standard wording and not real information about the firm's risks. This could weaken the link between risk-word density and volatility, but it does not by itself predict lower volatility.

H2. If Item 1A length and the other controls are held fixed, RiskDensity is associated with post-filing volatility: ($\beta_{\text{dens}} \neq 0$) in equation (3.3). The sign shows the direction of the association, but it does not tell us which channel is behind it. Density measures the share of selected words, not how specific the disclosure is, which is what Hope et al. (2016) study. Words such as "litigation" or "investigation" can also appear in descriptions of very specific events.

Sub-list decomposition. A second specification replaces RiskDensity with its two components:

$$\text{UncDensity}_{i,t} = \frac{n^{\text{unc}}_{i,t}}{\text{Words1A}_{i,t}}, \qquad \text{LitDensity}_{i,t} = \frac{n^{\text{lit}}_{i,t}}{\text{Words1A}_{i,t}}.$$

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \gamma' X_{i,t} + \beta_{\text{len}} \ln(\text{Words1A}_{i,t}) + \beta_{\text{unc}} \text{UncDensity}_{i,t} + \beta_{\text{lit}} \text{LitDensity}_{i,t} + \mu_{j(i,t)} + \tau_t + \varepsilon_{i,t}. \tag{3.5}$$

The uncertainty list captures uncertain and imprecise wording. The litigious list captures legal and regulatory language, which does not always point to an actual legal dispute or exposure (Loughran & McDonald, 2011). A Wald test of ($\beta_{\text{unc}} = \beta_{\text{lit}}$) checks whether the two densities are associated differently with volatility.

Ratio regressors. Both densities divide word counts by Item 1A length. Ratios can give misleading results if the role of the denominator is not handled carefully (Kronmal, 1993). The model includes log length to account for differences in section length, but this does not guarantee that the problem goes away. As a robustness check, I replace the densities with ($\ln(1+n^{\text{unc}})$) and ($\ln(1+n^{\text{lit}})$) and keep ln(Words1A) (Section 5.2). This model uses counts that are not divided by length. Comparing the two sets of results shows whether the findings depend on how risk language is measured.

3.5 H3 — Peer-relative omission gap

H3 asks whether the topics a firm *leaves out* of Item 1A, compared with its industry peers, help predict later volatility once length and density are accounted for. I call this measure the *omission gap*. One way to read it is as *strategic silence*: the firm says little about risks that its peers discuss. But other explanations would produce the same pattern. The firm may simply not face that risk, may use different words for it, or may write a shorter section for reasons unrelated to risk.

Construction. For each fiscal year t, I fit a TF–IDF model on the Item 1A texts filed before the first fiscal-year-t filing in the panel. This way, the vocabulary and weights are already known on the date of every filing I analyse (fiscal year 2010 has no earlier texts, so it has no omission gap). I use the standard `scikit-learn` Python library for the text vectorisation, with these settings: the text is lowercased; tokens must have at least two letters or digits; the library's English stop-word list is removed; only single words are used; a term must appear in at least five documents and in at most 95% of the training documents; raw term counts are used; inverse document frequency is smoothed; and each row is scaled to unit length. Let $\mathbf{x}_{i,t}$ be the TF–IDF row of firm i's fiscal-year-t Item 1A. The peers of a filing are the other panel firms in the same two-digit SIC industry (at the filing date). For each peer, I use the most recent Item 1A from fiscal year t or $t-1$ that was filed strictly before the focal filing, so no later information is used. With $P_{i,t}$ as this peer set,

$$\mathbf{p}_{i,t} = \sum_{k \in P_{i,t}} \mathbf{x}_k, \qquad \textsc{Omission}^{1A}_{i,t} \;=\; 1 \;-\; \frac{\sum_{w: x_{i,t,w} > 0} p_{i,t,w}}{\sum_{w} p_{i,t,w}}, \tag{3.5}$$

the share of peer TF–IDF weight that falls on terms the focal firm does not use. It lies between 0 and 1. Filings with fewer than five peers get no value.

Length and the sources of variation. The omission gap is closely tied to length, because a longer section naturally covers more of the peer vocabulary. In the H3 sample, its correlation with $\ln(\text{Words}^{1A})$ is -0.915 (Table 5.6). To deal with this, and to see where the variation in the gap comes from, I use three variants:

- a *length-invariant* omission gap. For each filing, I draw 20 random blocks of 3,000 consecutive tokens from its Item 1A (with a fixed seed), compute the gap on each block instead of the full text, and average the 20 values. Sections shorter than 3,000 tokens use the whole text;
- a *pooled-vocabulary* gap, using one TF–IDF model fitted on all years. This removes changes in vocabulary over time. Because it uses later documents, I use it only as a diagnostic;
- a breakdown of the year-on-year change in the gap into three parts: the change in the firm's own text (with peers held at $t-1$), the change in the peers' texts, and the change in who the peers are.

Specification. The H3 regression adds the omission gap to equation (3.3):

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \beta^{\text{len}} \ln(\text{Words}^{1A}_{i,t}) + \beta^{\text{dens}} \text{RiskDensity}_{i,t} + \beta^{\text{om}} \text{OMISSION}^{1A}_{i,t} + \mu + \tau_t + \varepsilon_{i,t}, \quad (3.6)$$

I estimate it with industry fixed effects ($\mu = \mu_j$) and with firm fixed effects ($\mu = \mu_i$). The controls are the same as in Section 3.2, so the H3 model differs from the H1/H2 model only by the omission gap.

H3 and the decision rule. *If length, density and the controls are held fixed, a larger omission gap is associated with higher post-filing volatility*, that is, $\beta^{\text{om}} > 0$. I judge the evidence with a fixed rule, which I wrote down in the project log before re-estimating the models:

1. The family of tests covers $h \in \{5, 10, 30, 90, 180\}$ days under both fixed-effect structures, which gives ten two-sided tests. I adjust the p-values with the Holm step-down procedure (Holm, 1979) at $\alpha = 0.05$.
2. The primary design is $h = 30$ days. Under a given fixed-effect structure, there is *positive evidence* if $\beta^{\text{om}} > 0$ with Holm-adjusted $p < 0.05$ *and* the length-invariant gap also has a positive coefficient with $p < 0.05$ in the same specification. *Negative evidence* is the mirror image.
3. H3 is *supported* if there is positive evidence under both structures. It is *partially supported* if there is positive evidence under one and no negative evidence under the other. It is *contradicted* if there is only negative evidence, and *not supported* in all other cases. Both fixed-effect structures are treated the same way.

I report the placebo test, the decomposition and the other robustness checks, but they do not change the verdict.

3.6 Estimation and inference

All regressions are estimated by ordinary least squares after absorbing the fixed effects (`linearmodels`). With industry and year fixed effects, the coefficients come from variation *between* firms in the same industry-year cell. I use firm fixed effects as a robustness check (Section 5.3); they rely only on year-to-year variation *within* a firm. The two designs answer different questions, and a coefficient from the industry fixed-effects model is not, on its own, evidence of a within-firm relationship. The analysis is predictive, not causal: the coefficients describe conditional associations between the disclosure and later volatility, in the spirit of Campbell et al. (2014).

Fiscal-year effects. Year fixed effects absorb shocks that hit all firms in a fiscal year. But fiscal years end in different months, so the filings for one fiscal year are spread over more than a year of calendar time (Section 2.3). A fiscal-year effect therefore does not remove market

shocks tied to calendar time. The volatility spike of spring 2020, for example, falls in the post-filing windows of firms with different fiscal-year labels, and in only part of the windows within a given label.

I run four robustness checks for this: the market (SPY) volatility over the same window as a control, filing-month × fiscal-year fixed effects, fixed effects for the calendar month of the filing in place of the fiscal-year effects, and a sample of firms with a December fiscal-year end, where fiscal and calendar time coincide. I also use idiosyncratic volatility as an extra dependent variable. It is the residual standard deviation from a market-model regression inside the window.

Anticipated volatility. Item 1A is written when management already knows current conditions, so expected volatility could shape the wording. Pre-filing volatility, which is public at filing, captures the part of this that is visible in prices. It does not rule out that management expects future volatility beyond that, so I read all results as associations.

Standard errors. Standard errors are clustered two-way by firm and fiscal year (Cameron et al., 2011; Petersen, 2009). Firm clustering allows residuals of the same firm to be correlated across years. Fiscal-year clustering allows residuals of different firms with the same fiscal-year label to be correlated. It does not fully capture calendar-time dependence that crosses fiscal-year labels.

With only 15 fiscal-year clusters, conventional cluster-robust tests may reject true null hypotheses too often (Cameron et al., 2008). For the text coefficients, I therefore also report wild cluster-bootstrap p-values. The bootstrap uses a studentized statistic, imposes the null, partials out the relevant fixed effects, draws weights at the fiscal-year level from Webb's (Webb, 2023) six-point distribution, and uses 9,999 replications. This is a one-way fiscal-year bootstrap: it does not by itself reproduce the full firm-and-year dependence structure. As a sensitivity check, I therefore also report p-values based on one-way firm clustering.

Dynamics and Nickell bias. Adding a lagged dependent variable to a firm fixed-effects model makes the transformed lag correlated with the transformed error. In the standard stationary autoregressive model, this biases the lag coefficient downward by an amount of order $(1/T)$, and it can also affect the other coefficients (Nickell, 1981). With at most 15 annual observations per firm, this is a concern. The main firm fixed-effects specifications use pre-filing volatility to control for earlier risk, and I report specifications with the lagged dependent variable for comparison. Pre-filing volatility is measured before the outcome window and differs from the lagged dependent variable. However, its timing alone does not guarantee exogeneity or remove every source of fixed-effects bias. A separate concern is non-stationarity, which a short time dimension and a large number of firms do not rule out by themselves. Section 5.5 therefore reports panel unit-root tests, with their results interpreted in light of the short time dimension and the tests' assumptions.

Outliers. Accounting ratios, book-to-market and the past return are winsorised as described in Section 3.2. The text variables are not winsorised: $\ln(\text{Words}^{1A})$ ranges from 7.41 to 10.48 between the 1st and 99th percentiles, and RiskDensity has a maximum of 0.0772 (Table 4.1).

Sample. The 30-day estimation sample has 5,782 firm-years from 512 firms; the same firm-years enter the 365-day models. Table A.1 in Appendix A shows how the sample is built, from the SEC ticker list down to this sample.

Limitation: survivorship and look-ahead. The universe is the first 1,000 rows of the SEC ticker list as stored in October 2026, which is ordered roughly by current size. Firms that were delisted, acquired or failed during 2010–2024 are missing, and firms that are large today are over-represented in the early years: the panel grows from 152 firm-years in fiscal year 2010 to over 500 from 2020 (Table A.2). Firms in distress are likely to be riskier, so the results describe large firms that survived. The sample rules themselves only look backwards: the liquidity requirement (Section 2.1) uses prices before the filing.

4. Results

This chapter presents the descriptive statistics and the tests of H1, H2 and H3 at the main 30-day horizon.

4.1 Descriptive statistics

Table 4.1 summarises the 30-day estimation sample of 5,782 firm-years. The median annualised 30-day post-filing volatility is 0.243 (mean 0.317). The median Item 1A has 8,700 tokens, but a few sections are much longer (the 99th percentile is 35,579). The mean RiskDensity is 0.0511, which means that about one token in twenty is an uncertainty or litigious word. Uncertainty words (mean density 0.0346) are about twice as common as litigious words (0.0165).

Table 4.1: Summary statistics (30-day estimation sample)

	Mean	SD	P1	P25	Median	P75	P99
$\sigma^{[30]}_{i,t+1}$	0.317	0.269	0.089	0.176	0.243	0.346	1.465
$\ln \sigma^{[30]}_{i,t+1}$	-1.347	0.579	-2.423	-1.736	-1.416	-1.063	0.382
$\ln \sigma^{\text{pre}}_{i,t}$	-1.294	0.433	-2.181	-1.597	-1.335	-1.042	-0.103
ln(Market cap)	23.591	1.436	20.252	22.737	23.548	24.451	26.983
Leverage	0.612	0.219	0.128	0.472	0.610	0.737	1.298
ROA	0.064	0.089	-0.319	0.028	0.063	0.107	0.294
Asset growth	0.117	0.247	-0.241	0.003	0.060	0.144	1.540
Book-to-market	0.299	0.255	-0.122	0.116	0.243	0.423	1.265
Past 12-month return	0.135	0.285	-0.753	-0.020	0.144	0.289	1.003
Current ratio	1.955	1.477	0.379	1.052	1.473	2.346	8.869
OCF/assets	0.112	0.079	-0.196	0.069	0.106	0.153	0.324
Words^{1A}	10,053	6,990	1,652	5,504	8,700	12,253	35,579
$\ln(\text{Words}^{1A})$	9.012	0.649	7.410	8.613	9.071	9.414	10.479
RiskDensity	0.0511	0.0065	0.0358	0.0471	0.0508	0.0552	0.0673
UncDensity	0.0346	0.0049	0.0225	0.0317	0.0346	0.0374	0.0472
LitDensity	0.0165	0.0046	0.0064	0.0134	0.0162	0.0194	0.0280
Observations	5,782						

Notes. $\sigma^{[30]}_{i,t+1}$: annualised standard deviation of daily log returns in the 30-calendar-day window starting two trading days after the 10-K filing day. σ^{pre}: the same over the 60 trading days before filing. Financial ratios, book-to-market and the past return are winsorised at the 1st/99th percentiles of this sample; the text variables are not winsorised. The maximum RiskDensity is 0.0772.

Figure 4.1 shows the median 30-day post-filing volatility by fiscal year, together with the inter-quartile range. Volatility is highest for fiscal year 2019. Those 10-Ks were filed in early 2020, just before and during the market turmoil of spring 2020. The fiscal-year fixed effects remove such common shocks as long as they affect all filings of the same fiscal year.

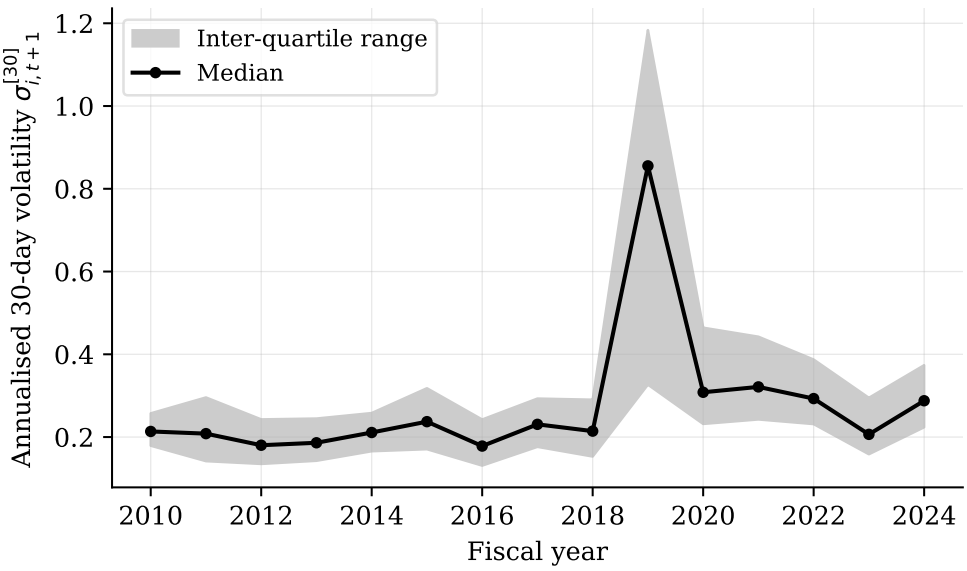


Figure 4.1: Median annualised 30-day post-filing volatility by fiscal year, with the inter-quartile range (shaded), 30-day estimation sample.

Table 4.2 shows the pairwise correlations. Length and RiskDensity are negatively correlated ($r = -0.30$): longer sections contain a smaller share of dictionary risk words, mostly uncertainty words ($r = -0.44$). The two sub-densities are almost uncorrelated ($r = -0.07$). Pre-filing and post-filing volatility are correlated at 0.53, which is why pre-filing volatility is the most important control.

Table 4.2: Pearson correlations (30-day estimation sample)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) $\ln \sigma^{[30]}_{i,t+1}$													
(2) $\ln \sigma^{\text{pre}}_{i,t}$	0.53												
(3) ln(Market cap)	-0.17	-0.29											
(4) Leverage	-0.07	-0.12	0.11										
(5) ROA	-0.23	-0.30	0.25	-0.04									
(6) Asset growth	0.15	0.15	-0.04	-0.12	-0.03								
(7) Book-to-market	0.04	0.03	-0.25	-0.30	-0.25	-0.06							
(8) Past 12-month return	-0.08	-0.16	0.08	0.00	0.10	0.10	-0.28						
(9) Current ratio	0.18	0.26	-0.19	-0.50	-0.06	0.19	-0.10	0.04					
(10) OCF/assets	-0.16	-0.17	0.25	-0.00	0.76	-0.07	-0.32	0.15	-0.07				
(11) ln(Words^{1A})	0.31	0.36	-0.02	-0.04	-0.28	0.16	-0.00	-0.01	0.17	-0.19			
(12) RiskDensity	-0.11	-0.17	0.02	-0.01	0.06	-0.07	-0.04	0.02	-0.08	-0.02	-0.30		
(13) UncDensity	-0.10	-0.10	-0.14	-0.06	0.14	-0.05	-0.04	0.03	-0.00	0.08	-0.44	0.71	
(14) LitDensity	-0.06	-0.13	0.18	0.05	-0.08	-0.05	-0.02	-0.00	-0.10	-0.12	0.05	0.65	-0.07

Notes. N as in Table 4.1.

4.2 Length and risk-word density (H1 and H2)

Table 4.3 shows five nested specifications estimated on the same sample: controls only (1), length (2), density (3), length and density together (4), and the split into the two sub-lists

(5).

H1. In the joint specification (column 4), a longer Item 1A goes together with higher post-filing volatility: $\hat{\beta}^{\text{len}} = +0.053$ ($t = +4.51$). The coefficient is similar in columns 2 and 5, and it stays significant under a wild cluster bootstrap by year ($p = 0.001$; Table 5.3). **H1 is supported** in the industry fixed-effects design. However, Section 5.3 shows that this link does not hold within firms over time.

H2. When length is held fixed, the composite RiskDensity has a negative coefficient ($\hat{\beta}^{\text{dens}} = -1.845$, $t = -1.70$, $p = 0.090$; wild bootstrap $p = 0.092$). Without the length control (column 3) it is -3.408 ($t = -3.33$). Length and density are negatively correlated, so part of the effect in column 3 is really a length effect. H2 is a two-sided test, and the composite coefficient is not significantly different from zero at the 5% level. So **H2 is not supported at the 30-day horizon**. At longer horizons the picture changes: from 90 days on, the composite coefficient is negative and significant at the 5% level ($t = -3.33$ at 365 days; Table 4.5).

Sub-lists. In column 5, uncertainty density is not related to volatility ($\hat{\beta}^{\text{unc}} = -0.545$, $t = -0.31$), while litigious density has a negative coefficient ($\hat{\beta}^{\text{lit}} = -3.018$, $t = -2.90$; wild bootstrap $p = 0.005$). The Wald test cannot reject that the two coefficients are equal ($p = 0.179$), so we cannot say that the two sub-lists really behave differently. Still, the negative sign of the composite comes mainly from the litigious words. This fits the boilerplate channel of Section 3.4 better than the risk-content channel, but the design cannot show which feature of litigious language is responsible.

Magnitudes. Table 4.4 puts the coefficients on a common scale. A one pooled standard deviation increase in $\ln(\text{Words}^{1A})$ (0.65 log points) goes together with 3.8% higher 30-day volatility. A one pooled standard deviation increase in LitDensity goes together with 1.4% lower volatility, and for the composite RiskDensity the figure is -1.2%. Changes within a single firm are much smaller (one within-firm standard deviation of length is 0.27 log points), so the same coefficients imply 1.6% and -0.6% for within-firm changes. At the median volatility of 0.243, a 3.8% change is about 0.9 percentage points of annualised volatility. The text variables explain very little additional variance: the within-R^2 only rises from 0.363 with controls alone to 0.368 (Table 4.3).

365-day horizon. Table 4.5 repeats the five columns with the 365-day window on the same firm-years. Length is still positive ($+0.084$, $t = +8.64$ in column 5), LitDensity is still negative (-4.709, $t = -5.38$), and UncDensity is not significant. At this horizon the Wald test does reject that the two sub-list coefficients are equal ($p < 0.001$).

Table 4.3: Item 1A length, risk-word density and post-filing volatility (30-day window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\text{Words}^{1A})$		+0.061***		+0.053***	+0.058***
		(+5.40)		(+4.51)	(+4.54)
RiskDensity			−3.408***	−1.845*	
			(−3.33)	(−1.70)	
UncDensity					−0.545
					(−0.31)
LitDensity					−3.018***
					(−2.90)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.614***	+0.588***	+0.603***	+0.585***	+0.584***
	(+10.88)	(+9.96)	(+10.27)	(+9.78)	(+9.85)
ln(Market cap)	−0.026***	−0.028***	−0.027***	−0.028***	−0.027***
	(−3.13)	(−3.35)	(−3.17)	(−3.35)	(−3.08)
Leverage	+0.038	+0.030	+0.030	+0.026	+0.027
	(+1.22)	(+0.96)	(+0.94)	(+0.84)	(+0.86)
ROA	−0.337***	−0.263***	−0.321***	−0.263***	−0.264***
	(−3.47)	(−2.71)	(−3.29)	(−2.73)	(−2.74)
Asset growth	+0.136**	+0.122**	+0.131**	+0.121**	+0.120**
	(+2.32)	(+2.11)	(+2.27)	(+2.10)	(+2.06)
Book-to-market	+0.021	+0.003	+0.015	+0.002	+0.003
	(+0.69)	(+0.11)	(+0.47)	(+0.06)	(+0.11)
Past 12-month return	+0.023	+0.015	+0.022	+0.015	+0.015
	(+0.51)	(+0.32)	(+0.48)	(+0.33)	(+0.32)
Current ratio	+0.022***	+0.018***	+0.020***	+0.017***	+0.017***
	(+4.49)	(+3.62)	(+4.02)	(+3.48)	(+3.47)
OCF/assets	−0.064	−0.096	−0.106	−0.114	−0.122
	(−0.38)	(−0.56)	(−0.61)	(−0.66)	(−0.72)
Industry and year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,782	5,782	5,782	5,782	5,782
Firms	512	512	512	512	512
Within R^2	0.363	0.367	0.365	0.368	0.368
Δ within R^2 vs (1)	0.0000	0.0042	0.0018	0.0047	0.0049
Wald $\beta^{\text{unc}} = \beta^{\text{lit}}$, p					0.179

Notes. Dependent variable $\ln \sigma_{i,t+1}^{[30]}$. All columns use the same sample and the common control vector. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. Within R^2: share of the variance left after the fixed effects that the regressors explain. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4.4: Economic magnitude of the text coefficients (30-day window)

FE	Variable	SD pooled	SD within firm	% per pooled SD	% per within SD
Industry + year	$\ln(\text{Words}^{1A})$	0.6488	0.2748	+3.83	+1.60
Industry + year	UncDensity	0.0049	0.0022	-0.27	-0.12
Industry + year	LitDensity	0.0046	0.0020	-1.37	-0.61
Firm + year	$\ln(\text{Words}^{1A})$	0.6488	0.2748	+0.56	+0.23
Firm + year	UncDensity	0.0049	0.0022	-1.16	-0.52
Firm + year	LitDensity	0.0046	0.0020	-1.42	-0.63
Industry + year	RiskDensity (col. 4)	0.0065	0.0030	-1.19	-0.55
Firm + year	RiskDensity (col. 4)	0.0065	0.0030	-1.75	-0.81

Notes. Change in annualised 30-day volatility, $100\,(e^{\hat{\beta}\,\Delta x} - 1)$, for an increase of one pooled standard deviation or one within-firm standard deviation (deviation from the firm mean) of the variable, using the coefficients of Table 4.3 (industry FE) and Table 5.4 (firm FE).

Table 4.5: Item 1A length, risk-word density and post-filing volatility (365-day window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\text{Words}^{1A})$		+0.086***		+0.076***	+0.084***
		(+9.81)		(+7.95)	(+8.64)
RiskDensity			−4.743***	−2.526***	
			(−7.08)	(−3.33)	
UncDensity					−0.106
					(−0.11)
LitDensity					−4.709***
					(−5.38)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.541***	+0.505***	+0.525***	+0.501***	+0.498***
	(+22.22)	(+21.05)	(+20.89)	(+20.58)	(+20.48)
ln(Market cap)	−0.037***	−0.039***	−0.038***	−0.040***	−0.037***
	(−6.83)	(−7.14)	(−7.32)	(−7.38)	(−7.06)
Leverage	+0.005	−0.006	−0.006	−0.010	−0.010
	(+0.19)	(−0.20)	(−0.19)	(−0.36)	(−0.34)
ROA	−0.503***	−0.398***	−0.481***	−0.398***	−0.401***
	(−6.10)	(−5.14)	(−6.19)	(−5.27)	(−5.41)
Asset growth	+0.077***	+0.058***	+0.070***	+0.057***	+0.054***
	(+5.70)	(+4.42)	(+5.22)	(+4.27)	(+4.11)
Book-to-market	−0.006	−0.031	−0.015	−0.033	−0.030
	(−0.16)	(−0.90)	(−0.43)	(−0.95)	(−0.87)
Past 12-month return	+0.075**	+0.064*	+0.074**	+0.064*	+0.064*
	(+2.27)	(+1.95)	(+2.19)	(+1.95)	(+1.95)
Current ratio	+0.019***	+0.013***	+0.017***	+0.013***	+0.013***
	(+4.14)	(+3.18)	(+3.68)	(+3.03)	(+2.99)
OCF/assets	+0.075	+0.031	+0.017	+0.005	−0.009
	(+0.63)	(+0.27)	(+0.15)	(+0.04)	(−0.08)
Industry and year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,782	5,782	5,782	5,782	5,782
Firms	512	512	512	512	512
Within R^2	0.575	0.591	0.582	0.593	0.594
Δ within R^2 vs (1)	0.0000	0.0159	0.0066	0.0175	0.0190
Wald $\beta^{\text{unc}} = \beta^{\text{lit}}$, p					0.000

Notes. Dependent variable $\ln \sigma_{i,t+1}^{[365]}$. All columns use the same sample and the common control vector. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. Within R^2: share of the variance left after the fixed effects that the regressors explain. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.3 The omission gap (H3)

Table 4.6 shows the H3 specification at six horizons and with both fixed-effect structures, for the omission gap and for its length-invariant version.

Table 4.6: H3: omission gap and post-filing volatility

	5 days	10 days	30 days	90 days	180 days	365 days
Omission^{1A}, Industry + year FE						
$\hat{\beta}$	−0.015	−0.032	+0.081	+0.238	+0.333	+0.206
	(−0.03)	(−0.10)	(+0.37)	(+1.21)	(+1.55)	(+1.17)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Omission^{1A}, Firm + year FE						
$\hat{\beta}$	+0.418	+0.414	+0.475	+0.639	+0.613	+0.274
	(+0.67)	(+0.49)	(+0.70)	(+1.09)	(+1.17)	(+0.61)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Length-invariant omission (3,000-token blocks), Industry + year FE						
$\hat{\beta}$	+0.542	+0.548**	+0.487	+0.452*	+0.443*	+0.285*
	(+1.32)	(+1.97)	(+1.60)	(+1.82)	(+1.90)	(+1.80)
Holm p	1.000	0.491	0.773	0.546	0.516	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Length-invariant omission (3,000-token blocks), Firm + year FE						
$\hat{\beta}$	+0.548	+0.679	+0.351	+0.433	+0.424	+0.097
	(+0.90)	(+1.08)	(+0.58)	(+0.82)	(+1.12)	(+0.34)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659

Notes. Each cell is a separate regression of $\ln \sigma^{[h]}_{i,t+1}$ on the omission measure, $\ln(\text{Words}^{1A})$, RiskDensity and the common controls. Holm p: Holm-adjusted over the ten tests of each measure at $h \in \{5,10,30,90,180\}$ under both FE structures (365 days is outside the family). Two-way clustered t-statistics. Verdict under the decision rule of Section 3.5: not supported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

At the 30-day horizon the omission coefficient is +0.081 (t = +0.37) with industry fixed effects and +0.475 (t = +0.70) with firm fixed effects. None of the ten coefficients in the family of tests is significant, even before the Holm adjustment, and all Holm-adjusted p-values equal one. The length-invariant gap tells the same story (+0.487, t = +1.60, and +0.351, t = +0.58). Following the decision rule of Section 3.5, **H3 is not supported**: once length, density and the controls are held fixed, the omission gap is not related to post-filing volatility. Chapter 5 shows that no variant of the measure changes this conclusion.

5. Robustness

This chapter checks the results of Chapter 4 against other horizons, specifications, samples and methods of inference, and examines the omission measure of H3 in detail.

5.1 Alternative horizons

Table 5.1 re-estimates the composite and decomposition specifications with volatility over 5, 10, 30, 90, 180 and 365 days. The length coefficient is positive and significant at the 1% level at every horizon (between +0.058 and +0.101 in the decomposition). LitDensity is negative and significant from 30 days on (t between −2.90 and −5.38), but not at 5 or 10 days, where the windows contain few returns. UncDensity is never significant. The composite RiskDensity is significant at the 5% level from 90 days on. Figure 5.1 plots the length and LitDensity coefficients. The coefficients are not compared formally across horizons, because the windows overlap and the estimates are correlated.

Table 5.1: Text coefficients at alternative horizons

	5 days	10 days	30 days	90 days	180 days	365 days
Decomposition specification (column 5)						
ln(Words^{1A})	+0.101***	+0.071***	+0.058***	+0.070***	+0.076***	+0.084***
	(+3.70)	(+4.48)	(+4.54)	(+7.60)	(+7.25)	(+8.64)
UncDensity	−1.353	+0.009	−0.545	+0.442	+0.544	−0.106
	(−0.39)	(+0.00)	(−0.31)	(+0.30)	(+0.49)	(−0.11)
LitDensity	+1.205	−2.017	−3.018***	−4.767***	−4.556***	−4.709***
	(+0.46)	(−1.08)	(−2.90)	(−4.54)	(−5.09)	(−5.38)
Composite specification (column 4)						
ln(Words^{1A})	+0.105***	+0.067***	+0.053***	+0.060***	+0.066***	+0.076***
	(+4.14)	(+4.76)	(+4.51)	(+7.48)	(+6.67)	(+7.95)
RiskDensity	−0.025	−1.056	−1.845*	−2.296**	−2.137**	−2.526***
	(−0.01)	(−0.69)	(−1.70)	(−2.16)	(−2.55)	(−3.33)
Observations	5,300	5,775	5,782	5,782	5,782	5,782
Within R^2 (col. 5)	0.170	0.291	0.368	0.443	0.522	0.594

Notes. Each column re-estimates the specification with $\ln \sigma^{[h]}_{i,t+1}$ as the dependent variable; common controls, industry and year FE. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

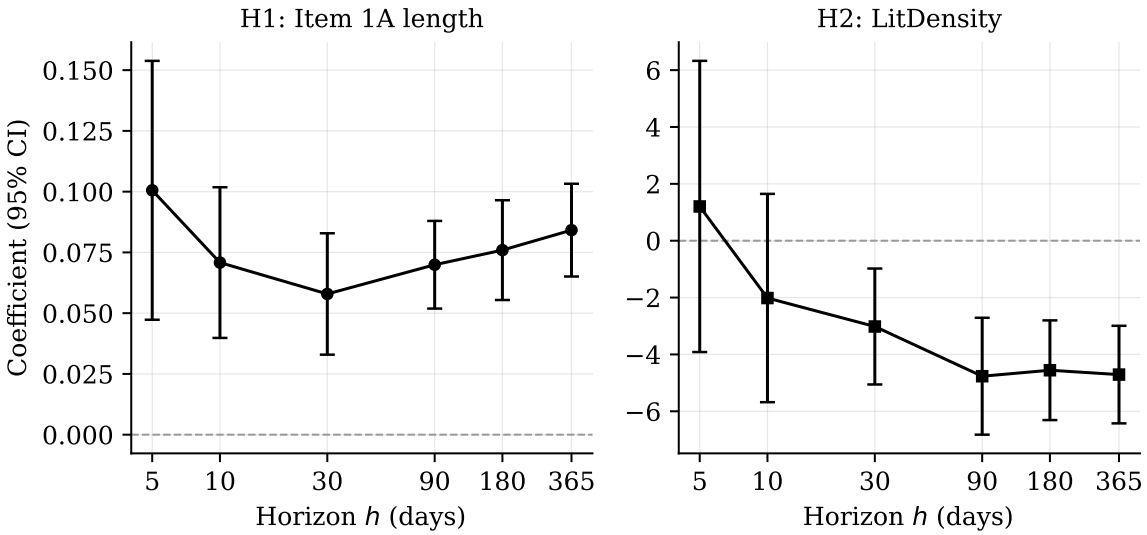


Figure 5.1: Length (left) and LitDensity (right) coefficients with 95% confidence intervals at the six horizons (decomposition specification, industry and year fixed effects; horizons on a log scale).

5.2 Robustness of the H1 and H2 estimates

Table 5.2 reports the decomposition specification under alternative choices, one change per row (Table A.8 in Appendix A repeats it at 365 days).

The length result is highly stable: the coefficient remains positive and statistically significant across the alternative market, time, industry, sample, timing, and inference specifications reported in Table 5.2. The LitDensity result is more sensitive. Its significance weakens when firm-years with imputed balance-sheet observations are removed, in the December-only sample, and when counts replace density ratios; the remaining checks leave the substantive conclusion unchanged. Thus H1 is robust across the tested specifications, whereas the evidence for H2 depends more on the measurement and sample definition.

Table 5.2: Robustness of the H1/H2 decomposition (30-day window)

	$\ln(\text{Words}^{1A})$	UncDensity	LitDensity	N
Baseline (Table 4.3, col. 5)	+0.058*** (+4.54)	−0.545 (−0.31)	−3.018*** (−2.90)	5,782
+ market volatility in window	+0.067*** (+5.75)	+0.139 (+0.11)	−3.399*** (−4.05)	5,782
Windows with an earnings release dropped	+0.056*** (+4.42)	−0.585 (−0.34)	−3.048*** (−2.96)	5,703
Volatility without earnings-release days	+0.057*** (+4.48)	−0.511 (−0.29)	−3.093*** (−3.03)	5,782
Idiosyncratic volatility as DV	+0.089*** (+6.02)	−2.589 (−1.57)	−3.308*** (−2.61)	5,782
Filing-month × year FE	+0.053*** (+4.61)	+0.020 (+0.01)	−2.235** (−2.37)	5,782
Calendar year-month of filing FE (instead of fiscal year)	+0.057*** (+4.94)	+0.217 (+0.16)	−2.329*** (−2.62)	5,782
Firms with a December fiscal-year end only	+0.060*** (+4.64)	+1.057 (+0.63)	−1.906* (−1.96)	4,200
SIC3 + year FE	+0.065*** (+4.16)	−0.432 (−0.26)	−3.542** (−2.49)	5,782
FF48 + year FE	+0.066*** (+4.89)	−0.944 (−0.56)	−4.052*** (−3.64)	5,782
Today's SIC2 + year FE	+0.058*** (+4.73)	−0.523 (−0.30)	−2.951*** (−2.76)	5,782
Excluding utilities (49) and health services (80)	+0.064*** (+4.91)	−1.406 (−0.70)	−3.005*** (−2.58)	5,112
Excluding weighted-average share counts	+0.047*** (+3.55)	−0.339 (−0.19)	−3.953*** (−3.57)	5,392
Excluding imputed balance-sheet values	+0.078*** (+4.87)	−0.760 (−0.43)	−1.761 (−1.05)	3,820
Balanced panel, FY 2012–2024	+0.081*** (+4.80)	−0.128 (−0.06)	−3.405** (−2.47)	3,367
Firms with ≥ 3 years	+0.057*** (+4.57)	−0.524 (−0.30)	−2.997*** (−2.88)	5,765
Firms with ≥ 5 years	+0.055*** (+4.41)	−0.649 (−0.37)	−3.072*** (−3.01)	5,613
Firms with ≥ 7 years	+0.056*** (+4.31)	−0.702 (−0.39)	−3.181*** (−2.97)	5,432
+ lagged dependent variable	+0.055*** (+4.02)	−0.516 (−0.29)	−2.866*** (−2.77)	5,782
SE clustered by firm only	+0.058*** (+4.87)	−0.545 (−0.40)	−3.018** (−2.19)	5,782
SE clustered by year only	+0.058*** (+4.97)	−0.545 (−0.33)	−3.018*** (−3.90)	5,782
Counts: ln(Words), $\ln(1+n^{\text{unc}})$, $\ln(1+n^{\text{lit}})$	−0.050* (−1.76)	+0.098** (+2.47)	−0.001 (−0.09)	5,855
Text of year $t-1$ instead of t	+0.057*** (+4.42)	+0.175 (+0.11)	−2.449* (−1.95)	5,345
Level + change: Level $t-1$	+0.063*** (+4.77)	+0.009 (+0.01)	−2.696** (−2.43)	5,289
Level + change: Change $t-1 \to t$	+0.080 (+1.47)	−2.850 (−0.99)	−1.075 (−0.25)	5,289
Post-2019 interaction: FY $\leq$ 2019 (main effect)	+0.046*** (+4.04)	−1.756 (−0.72)	−2.108 (−1.47)	5,782
Post-2019 interaction: × FY $\geq$ 2020	+0.039 (+1.42)	+3.705 (+1.02)	−3.363 (−1.09)	5,782

Notes. Each row is a separate regression of the column-(5) specification with the change described; common controls, industry (historical SIC2) and year FE and two-way clustered standard errors unless stated. The count-based row is not directly comparable to the baseline, because the log-count variables are observed for more firm-years than the density ratios; the raw counts are still defined for short or weakly parsed Item 1A sections, while the density measures are missing there. A like-for-like comparison therefore requires restricting both specifications to the same sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5.3: Inference with few year clusters (30-day window)

		Coef.	p-value		
FE	Variable		Two-way	Firm	Wild bootstrap (year)
Industry + year	$\ln(\text{Words}^{1A})$ (4)	+0.053	0.000		0.001
Industry + year	RiskDensity (4)	−1.845	0.090		0.092
Industry + year	$\ln(\text{Words}^{1A})$ (5)	+0.058	0.000	0.000	0.001
Industry + year	UncDensity (5)	−0.545	0.757	0.692	0.767
Industry + year	LitDensity (5)	−3.018	0.004	0.028	0.005
Firm + year	$\ln(\text{Words}^{1A})$ (4)	+0.008	0.688		0.687
Firm + year	RiskDensity (4)	−2.727	0.189		0.212
Firm + year	$\ln(\text{Words}^{1A})$ (5)	+0.009	0.654		0.662
Firm + year	UncDensity (5)	−2.372	0.212		0.204
Firm + year	LitDensity (5)	−3.132	0.308		0.349

Notes. Wild cluster bootstrap by fiscal year (15 clusters), 9,999 draws, Webb six-point weights, null imposed, fixed effects partialled out; (4)/(5) = column of Table 4.3. Firm: one-way firm-clustered p (industry FE, column 5).

5.3 Firm fixed effects

The headline models compare firms within an industry and year. Replacing the industry effects by firm effects asks a different question: whether a firm whose Item 1A becomes longer or denser than usual has higher volatility than usual. Table 5.4 reports the answer, and Figure 5.2 compares the coefficients.

Table 5.4: Industry versus firm fixed effects (30-day window)

	(1) Industry + year	(2) Firm + year	(3) Firm + year, lagged DV
$\ln(\text{Words}^{1A})$	+0.058***	+0.009	+0.015
	(+4.54)	(+0.45)	(+0.65)
UncDensity	−0.545	−2.372	−2.492
	(−0.31)	(−1.25)	(−1.19)
LitDensity	−3.018***	−3.132	−3.327
	(−2.90)	(−1.02)	(−1.05)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.584***	+0.436***	+0.451***
	(+9.85)	(+6.53)	(+8.05)
$\ln \sigma_{i,t}^{[30]}$ (lagged)			−0.099
			(−1.46)
RiskDensity in column-(4) specification	−1.845*	−2.727	−2.883
	(−1.70)	(−1.31)	(−1.31)
Observations	5,782	5,782	5,782
Firms	512	512	512
Within R^2	0.368	0.115	0.125

Notes. All columns include the common control vector and use the same sample. Column (3) adds the lagged dependent variable, which is subject to Nickell bias under firm fixed effects. Standard errors clustered two-way by firm and fiscal year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Under firm fixed effects the length coefficient falls from +0.058 to +0.009 ($t = +0.45$), and no density coefficient is significant (LitDensity −3.132, $t = -1.02$). Adding the lagged dependent variable, whose coefficient is biased under firm fixed effects (Nickell, 1981), changes nothing (column 3). The associations in Chapter 4 are therefore between-firm patterns: firms with longer and more litigious risk-factor sections differ systematically in subsequent volatility from otherwise similar firms in the same industry and year, but year-to-year changes in a firm's own section do not track changes in its volatility. Two features of the data limit what the firm fixed-effects models can show. First, most of the variation in the text variables is between firms: only 18% of the variance of $\ln(\text{Words}^{1A})$ and 19% of that of LitDensity is within firms (Table 5.5), so the within estimates are imprecise; the confidence interval of the within-firm LitDensity coefficient contains the between-firm estimate. Second, Item 1A changes slowly from year to year in this sample: only 18% of the variance of log Item 1A length and 19% of LitDensity is within firms (Table 5.5). This is a finding from the panel, not a direct implication of the MD&A study of Brown and Tucker (2011) or the legal and borrowing evidence in Cazier et al. (2021).

Table 5.5: Within-firm and between-firm variation

	N	SD	SD within	Within share	Within share net of year
$\ln \sigma^{[30]}_{i,t+1}$	5,782	0.5794	0.4746	0.671	0.380
$\ln(\text{Words}^{1A})$	5,782	0.6488	0.2748	0.179	0.117
RiskDensity	5,782	0.0065	0.0030	0.213	0.207
UncDensity	5,782	0.0049	0.0022	0.204	0.202
LitDensity	5,782	0.0046	0.0020	0.194	0.179
Omission^{1A}	4,659	0.1001	0.0358	0.128	0.122

Notes. Within share: variance of the deviation from the firm mean over the total variance; the last column first removes fiscal-year means. 30-day estimation sample (omission: H3 sample).

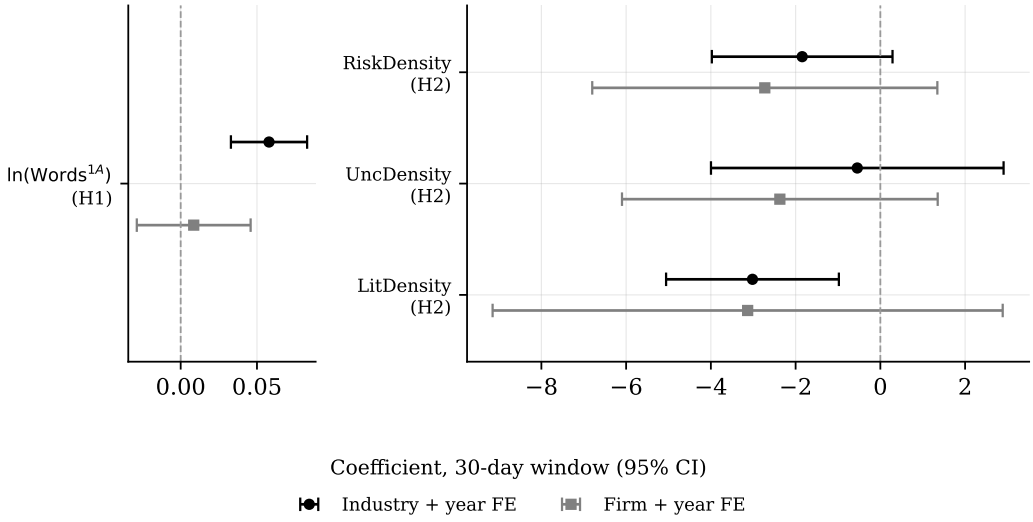


Figure 5.2: Text coefficients at the 30-day horizon with 95% confidence intervals, industry and year fixed effects (circles) versus firm and year fixed effects (squares). The length coefficient falls from +0.058 to +0.009; the density intervals under firm fixed effects are wide and include zero.

5.4 The omission measure (H3)

H3 is not supported in Chapter 4. This section reports three diagnostics that clarify the construction and interpretation of the measure. Table 5.6 summarises them.

Table 5.6: H3 diagnostics: placebo, sources of variation, and length

Placebo: random SIC2 cells within year, firm + year FE, 30 days	
Permutations	1,000
Observed t	0.70
Placebo t: mean / SD	0.24 / 0.93
Placebo t: 95th / 97.5th percentile	1.69 / 1.96
Empirical p (one-sided / two-sided)	0.315 / 0.466
Decomposition of ΔOmission (n = 4,802)	
Own-text change: share of variance	0.907
Peer-text change: share of variance	0.063
Peer-composition change: share of variance	0.026
Interaction: share of variance	0.078
Correlation with $\ln(Words^{1A})$*, H3 sample*	
Omission^{1A}	-0.915
N	4,659

Notes. Decomposition in the year-t vocabulary using fiscal-year documents: own $= \Omega(o_t, C_{t-1}) - \Omega(o_{t-1}, C_{t-1})$; peer text = change for the peers present in both years; composition = effect of entering/leaving peers; interaction = remainder.

Length. The omission gap is strongly related to length: $r = -0.915$ with $\ln(\text{Words}^{1A})$ in the H3 sample of 4,659 firm-years (Figure 5.3), indicating that the measure is closely tied to disclosure length.

Sources of variation. The decomposition in Table 5.6 attributes 91% of the year-on-year change to the firm's own text, 6% to peer-text changes, and 3% to peer composition.

Placebo. Firms were reassigned at random to industry cells within each fiscal year, keeping the cell sizes, the gap was recomputed and the firm fixed-effects model re-estimated, 1,000 times. The observed t-statistic of +0.70 lies in the middle of the placebo distribution: the empirical two-sided p-value is 0.466, and 2.4% of the placebo t-statistics exceed 1.96 (Figure 5.4). The real peer groups carry no more information about volatility than random ones.

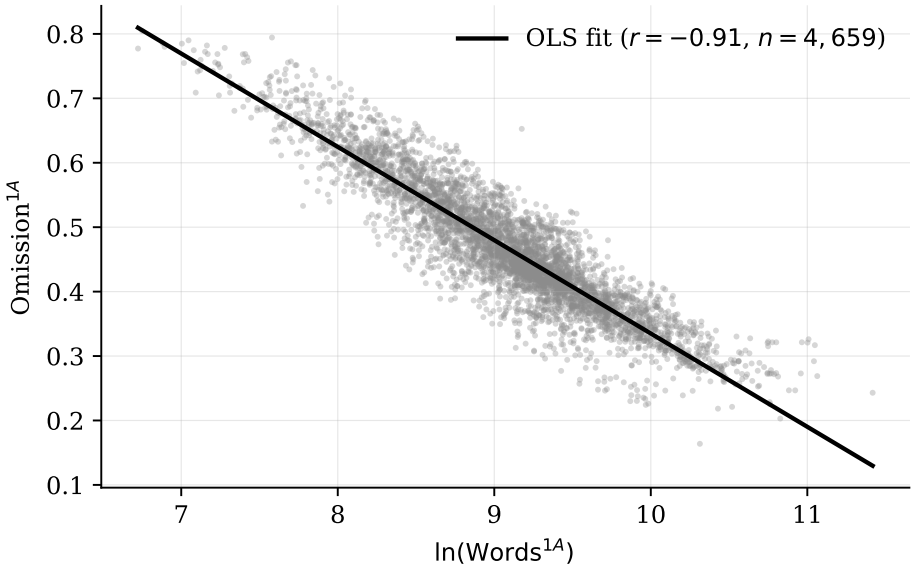


Figure 5.3: Omission gap against ln(Words^{1A}), one point per firm-year of the 30-day H3 sample, with the OLS fit.

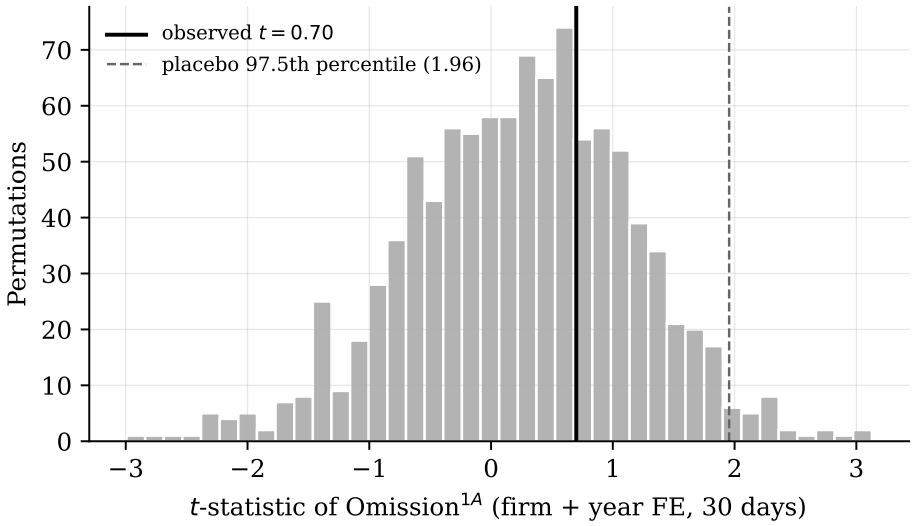


Figure 5.4: Distribution of the omission-gap t-statistic in 1,000 placebo assignments of firms to industry cells (firm and year fixed effects, 30-day window), with the observed value.

5.5 Persistence and unit roots

With at most 15 annual observations per firm, the dynamic panel raises two separate concerns: finite-T bias in specifications with a lagged dependent variable, and possible non-stationarity. The Nickell bias discussed in Section 3.6 does not establish that the second concern is negligible. Table 5.7 therefore reports two panel unit-root tests designed for short panels. Every variable is first demeaned by fiscal year; this removes a common additive year component but is not a general cure for cross-sectional dependence. The Harris–Tzavalis test (Harris & Tzavalis, 1999) uses the balanced panel of firms observed in all years 2012–2024 and tests the null that every firm's series has a unit root against a common autoregressive coefficient below one. The Im–Pesaran–Shin test (Im et al., 2003) averages firm-level augmented Dickey–Fuller statistics on the unbalanced panel; its null is that all firms have a unit root, and a rejection means that at least some firms' series are stationary, not that all are. Both procedures rely on their panel-specific innovation, dependence and ADF-specification assumptions. The simulated p-values reported here are adaptations calibrated to the observed sample dimensions, rather than values supplied by the cited papers.

Table 5.7: Panel unit-root tests

	Harris–Tzavalis (balanced, 2012–2024)			Im–Pesaran–Shin (1 lag)		
	N	$\hat{\rho}$	p	N	$W_{\bar{t}}$	p
$\ln \sigma^{[30]}_{i,t+1}$	379	-0.077	0.000	449	-16.47	0.000
$\ln \sigma^{\text{pre}}_{i,t}$	379	0.180	0.000	449	-16.29	0.000
ln(Market cap)	371	0.460	0.000	437	-3.76	0.000
Leverage	344	0.772	0.592	437	-3.67	0.000
ROA	358	0.419	0.000	433	-9.11	0.000
Asset growth	378	-0.052	0.000	449	-20.82	0.000
Book-to-market	362	0.516	0.000	430	-6.89	0.000
Past 12-month return	379	-0.108	0.000	449	-26.48	0.000
Current ratio	368	0.446	0.000	439	-10.10	0.000
OCF/assets	377	0.437	0.000	448	-9.49	0.000
ln(Words^{1A})	339	0.351	0.000	417	-14.07	0.000
RiskDensity	321	0.673	0.000	402	-4.40	0.000
UncDensity	321	0.659	0.000	402	-5.36	0.000
LitDensity	321	0.697	0.000	402	-1.92	0.027
Omission^{1A}	220	0.758	0.257	335	-1.29	0.099

Notes. Variables cross-sectionally demeaned by fiscal year. Harris–Tzavalis: within estimate of ρ on firms observed in every year 2012–2024 ($T = 13$); H_0: $\rho = 1$ for all firms; p from 2,000 simulated random-walk panels of the same N and T. IPS: mean of firm ADF t-statistics (intercept, one lag; firms with ≥ 8 consecutive years), standardised with simulated null moments; H_0: all firms have a unit root, H_1: some firms are stationary.

Both tests reject the unit-root null for volatility, the text variables and most controls. Two variables are exceptions. For leverage the Harris–Tzavalis test does not reject ($\hat{\rho} = 0.772$, $p = 0.592$), while the Im–Pesaran–Shin test does; for the omission gap neither test rejects at the 5% level ($\hat{\rho} = 0.758$, $p = 0.257$; IPS $p = 0.099$). Leverage is a control, and the omission gap enters only the H3 models, whose coefficient is insignificant. A ratio bounded between zero and one cannot follow a random walk forever, but over 13 years it can be persistent

enough that these tests lack power; the results are therefore read as showing high persistence for these two variables rather than as proof of stationarity.

Conclusion

This thesis asked whether three properties of the risk-factor section of the 10-K, its length, its share of uncertainty and litigious words, and what it leaves out relative to industry peers, are associated with stock-return volatility in the weeks after the filing. The tests use a panel of 5,782 firm-years of 512 large U.S. non-financial firms for fiscal years 2010–2024, built from SEC filings, and control for the volatility before the filing, firm size, valuation, past returns and accounting fundamentals.

Findings. *H1 is supported between firms.* Firms with longer Item 1A sections have higher 30-day post-filing volatility than otherwise similar firms in the same industry and year: one standard deviation of log length is associated with 3.8% higher volatility. The association holds at every horizon from 5 to 365 days, under alternative industry definitions, with market and idiosyncratic volatility, and with a wild cluster bootstrap. It does not hold within firms: when a firm's own section becomes longer, its volatility does not rise with it.

H2 is not supported at the 30-day horizon. Holding length fixed, the composite density of uncertainty and litigious words has a negative but insignificant coefficient at 30 days; it is negative and significant only at horizons of 90 days and longer. Of its two parts, the uncertainty density has no association with volatility at any horizon, while the litigious density is negative and significant from 30 days on, but this result weakens when firm-years with imputed balance-sheet values are excluded and disappears within firms. The negative sign is more in line with litigious vocabulary acting as standard cautionary language than with it signalling additional risk, but the evidence for H2 is weak.

H3 is not supported. The share of the peers' vocabulary that a firm does not use in its Item 1A is not associated with subsequent volatility under industry or firm fixed effects, after a Holm adjustment over horizons and fixed-effect structures, in a length-invariant version of the measure, or relative to 1,000 random peer assignments. The measure is dominated by section length, and its year-to-year variation comes mainly from the firm's own revisions. The data give no support for a strategic-silence interpretation of omissions.

Together, the results suggest that the volume of risk disclosure is the most robust textual correlate of post-filing volatility among the measures studied, and that it reflects persistent differences between firms rather than year-to-year changes in what a firm discloses. Once pre-filing volatility is controlled for, the text variables add little explanatory power: the within-industry-year R^2 rises from 0.363 to 0.368.

Limitations. The results are associations, not causal effects. The firm list was taken from the current SEC ranking, so firms that failed or were acquired during the sample period are missing and early years contain only firms that later became large. Industry peers are defined

by two-digit SIC codes, which group firms with different businesses; alternative industry definitions do not change the H1 and H2 results, but the omission measure depends on the peer definition. With at most 15 years per firm and little within-firm variation in the text variables, the firm fixed-effects estimates are imprecise. The text extraction was audited by two language models on a stratified sample of 200 sections and checked by hand on 51 of them; about 5–7% of sections are deficient, mostly at their boundaries and in early years, and the share of sections taken from exhibits rather than the main document was not recorded. The dictionary densities measure the vocabulary of a section, not its specificity.

Further research. The finding that length matters between but not within firms invites a closer look at what makes some firms' disclosures persistently longer, for example their business complexity or litigation history. Measures of disclosure content that do not depend on fixed word lists, and peer groups based on product descriptions rather than SIC codes (Hoberg & Phillips, 2016), could test the omission idea with less noise. A broader, point-in-time sample including firms that later left the market would show whether the results extend beyond large surviving firms.

References

Bao, Y., & Datta, A. (2014). Simultaneously discovering and quantifying risk types from textual risk disclosures. *Management Science*, *60*(6), 1371–1391. https://doi.org/10.1287/mnsc.2014.1930

Brown, S. V., & Tucker, J. W. (2011). Large-sample evidence on firms' year-over-year MD&A modifications. *Journal of Accounting Research*, *49*(2), 309–346.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, *90*(3), 414–427. https://doi.org/10.1162/rest.90.3.414

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, *29*(2), 238–249.

Campbell, J. L., Chen, H., Dhaliwal, D. S., Lu, H.-m., & Steele, L. B. (2014). The information content of mandatory risk factor disclosures in corporate filings. *Review of Accounting Studies*, *19*(1), 396–455.

Cazier, R. A., McMullin, J. L., & Treu, J. S. (2021). Are lengthy and boilerplate risk factor disclosures inadequate? an examination of judicial and regulatory assessments of risk factor language. *The Accounting Review*, *96*(4), 131–155.

Chychyla, R., & Kogan, A. (2015). Using XBRL to conduct a large-scale study of discrepancies between the accounting numbers in Compustat and SEC 10-K filings. *Journal of Information Systems*, *29*(1), 37–72. https://doi.org/10.2308/isys-50922

Dyer, T., Lang, M., & Stice-Lawrence, L. (2017). The evolution of 10-K textual disclosure: Evidence from Latent Dirichlet Allocation. *Journal of Accounting and Economics*, *64*(2–3), 221–245.

Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *The Journal of Finance*, *47*(2), 427–465. https://doi.org/10.1111/j.1540-6261.1992.tb04398.x

First Community Financial Corporation. (2012). Annual Report on Form 10-K for the Fiscal Year Ended December 31, 2011 [SEC EDGAR filing, accession no. 0001193125-12-106395]. https://www.sec.gov/Archives/edgar/data/763293/000119312512106395/d277464d10k.htm

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120.

Harris, R. D. F., & Tzavalis, E. (1999). Inference for unit roots in dynamic panels where the time dimension is fixed. *Journal of Econometrics*, *91*(2), 201–226. https://doi.org/10.1016/S0304-4076(98)00076-1

Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, *124*(5), 1423–1465. https://doi.org/10.1086/688176

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70.

Hope, O.-K., Hu, D., & Lu, H. (2016). The benefits of specific risk-factor disclosures. *Review of Accounting Studies*, *21*(4), 1005–1045.

Im, K. S., Pesaran, M. H., & Shin, Y. (2003). Testing for unit roots in heterogeneous panels. *Journal of Econometrics*, *115*(1), 53–74. https://doi.org/10.1016/S0304-4076(03)00092-7

Kogan, S., Levin, D., Routledge, B. R., Sagi, J. S., & Smith, N. A. (2009). Predicting risk from financial reports with regression. *Proceedings of Human Language Technologies: The 2009 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, 272–280. https://doi.org/10.3115/1620754.1620794

Kravet, T., & Muslu, V. (2013). Textual risk disclosures and investors' risk perceptions. *Review of Accounting Studies*, *18*(4), 1088–1122.

Kronmal, R. A. (1993). Spurious correlation and the fallacy of the ratio standard revisited. *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, *156*(3), 379–392. https://doi.org/10.2307/2983064

Li, F. (2008). Annual report readability, current earnings, and earnings persistence. *Journal of Accounting and Economics*, *45*(2–3), 221–247.

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, *66*(1), 35–65.

Loughran, T., & McDonald, B. (2016). Textual analysis in accounting and finance: A survey. *Journal of Accounting Research*, *54*(4), 1187–1230. https://doi.org/10.1111/1475-679X.12123

Loughran, T., & McDonald, B. (2025). Loughran–mcdonald master dictionary, 1993–2025 [Downloaded from the University of Notre Dame's Securities Research Initiative]. https://sraf.nd.edu/loughranmcdonald-master-dictionary/

Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, *49*(6), 1417–1426. https://doi.org/10.2307/1911408

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, *22*(1), 435–480.

Sonesta International Hotels Corporation. (2010). Annual Report on Form 10-K for the Fiscal Year Ended December 31, 2009 [SEC EDGAR filing, accession no. 0000091741-10-000005]. https://www.sec.gov/Archives/edgar/data/91741/000009174110000005/form10k.htm

United States. (2026a). 17 C.F.R. Section 232.303: Incorporation by reference [Code of Federal Regulations, Title 17, Part 232, paragraph (b)]. https://www.law.cornell.edu/cfr/text/17/232.303

United States. (2026b). 17 C.F.R. Section 240.12b-23: Incorporation by reference [Code of Federal Regulations, Title 17, Part 240]. https://www.law.cornell.edu/cfr/text/17/240/12b-23

U.S. Securities and Exchange Commission. (2009). Interactive Data to Improve Financial Reporting [Release No. 33-9002]. https://www.sec.gov/rules/final/2009/33-9002.pdf

U.S. Securities and Exchange Commission. (2014). Form 10-K [SEC Form 1673, revision 11/14. See General Instruction G(2) and Note 2]. https://www.sec.gov/files/2017-03/form10-k.pdf

U.S. Securities and Exchange Commission. (2016). Business and Financial Disclosure Required by Regulation S-K [Concept Release, Release Nos. 33-10064 and 34-77599]. https://www.sec.gov/files/rules/concept/2016/33-10064.pdf

Wal-Mart Stores, Inc. (2014). Annual Report on Form 10-K for the Fiscal Year Ended January 31, 2014 [SEC EDGAR filing, accession no. 0000104169-14-000019]. https://www.sec.gov/Archives/edgar/data/104169/000010416914000019/wmtform10-kx13114.htm

Webb, M. D. (2023). Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics*, *56*(3), 839–858. https://doi.org/10.1111/caje.12661

Appendices

A. Data Construction and Supplementary Tables

This appendix collects the tables on sample construction and data quality referred to in Chapters 2–5. Definitions, sources and timing of all variables are documented in the file `docs/data_dictionary.md` of the accompanying repository.

A.1 Sample construction and coverage

Table A.1: Sample construction

Stage	Firms	Firm-years
Universe (SEC ticker list, reconstructed with current metadata)		
SEC ticker list, first 1,000 rows	1,000	
− not an operating company	753	
− financial (SIC 6000-6999)	600	
− no fiscal-year end	599	
− < 5 years of 10-K XBRL data	550	
Frozen universe (548 firms) to panel		
≥ 5 years of 10-K XBRL data	548	8,000
transitions & dedup	548	7,958
drop earliest year	548	7,410
cap FY ≤ 2024	548	6,757
Panel to 30-day estimation sample		
valid 30d post-filing volatility	538	6,603
pre-filing liquidity rule	538	6,598
text from the panel's 10-K	538	6,592
Item 1A text (≥ 100 tokens)	528	6,387
non-missing pre_log_vol	528	6,387
non-missing log_mcap	522	6,253
non-missing leverage	521	6,083
non-missing roa	520	5,934
non-missing asset_growth	520	5,919
non-missing bm	520	5,919
non-missing ret_pre_12m	520	5,919
non-missing current_ratio	512	5,787
non-missing ocf_to_assets	512	5,782

Notes. The 1,000 firms are the first 1,000 rows of the SEC `company_tickers.json` list as stored in the repository (2026-10-12); the universe file holds the 548 firms that passed these filters in the original run. Each estimation-sample row applies one more requirement.

Table A.2: Text coverage by fiscal year

Fiscal year	Filings	Item 1A (%)	Item 7 (%)
2010	152	92.8	90.1
2011	313	94.2	92.0
2012	403	95.3	92.6
2013	424	95.5	93.4
2014	430	96.7	94.4
2015	442	95.9	94.1
2016	460	96.3	94.6
2017	473	96.0	94.5
2018	484	97.5	96.1
2019	496	97.2	95.8
2020	514	96.7	95.9
2021	532	97.4	96.6
2022	542	97.2	96.5
2023	547	97.6	96.9
2024	545	94.7	93.9
All	6,757	96.3	94.9

Notes. Share of panel firm-years whose stored section has at least 100 tokens.

Table A.3: Fiscal-year ends and filing dates in the estimation sample (30-day window)

Month	Fiscal year ends (%)	10-K filed (%)
Jan	5.0	2.8
Feb	0.0	66.5
Mar	2.4	8.7
Apr	1.2	0.6
May	1.7	2.3
Jun	5.4	1.6
Jul	1.6	1.5
Aug	1.2	5.2
Sep	5.2	1.6
Oct	2.7	1.4
Nov	1.0	5.0
Dec	72.6	2.7

Notes. Shares of the 5,782 firm-years. Fiscal-year end month: month of the period end, periods ending on 1–7 January counted to December. The median lag between period end and filing is 53 days (10th percentile 38, 90th percentile 59). 145 of the 512 firms have a fiscal year that does not end in December. 73.6% of the 10-Ks are filed in the calendar year after the fiscal year, and the filings of one fiscal year span a median of 390 calendar days; the filings fall in 169 distinct calendar months.

A.2 XBRL tags

Table A.4: XBRL tags: choices, locking losses and overlap

Metric	Locked tag	Firms	%
Net income	`ProfitLoss`	322	58.8
	`NetIncomeLoss`	223	40.7
	`IncomeLossFromContinuingOperations`	3	0.5
Operating cash flow	`NetCashProvidedByUsedInOperatingActivities`	530	96.7
	`NetCashProvidedByUsedInOperatingActivities` `ContinuingOperations`	18	3.3
Stockholders' equity	`StockholdersEquityIncluding` `PortionAttributableToNoncontrollingInterest`	315	58.0
	`StockholdersEquity`	228	42.0
Values set to missing by per-firm tag locking			
Stockholders' equity	637 of 7,900	171 firms with >1 tag	
Net income	38 of 7,813	209 firms with >1 tag	
Operating cash flow	0 of 7,956	217 firms with >1 tag	
Paired tags reported together: |difference| / total assets (pp)			
Net income	n = 3,838; median 0.043; p95 1.48; p99 3.56; max 37.4	>1%: 303	7.9
Operating cash flow	n = 417; median 0.035; p95 1.80; p99 4.19; max 6.5	>1%: 50	12.0

Notes. Net-income pair: `ProfitLoss` (consolidated) vs `NetIncomeLoss` (parent); their difference is the non-controlling interest, and the panel uses the consolidated figure. Operating-cash-flow pair: total vs continuing operations; the tags are not strictly equivalent, because they differ when discontinued operations have cash flows. Equity: 315 firms are locked on the tag including NCI and 228 on the parent-only tag; computed liabilities (total assets minus equity) contain NCI for the latter, while book-to-market uses parent equity first. ">1%": differences above 1% of total assets. Locking: for metrics with non-equivalent tags, each firm keeps its most frequent tag.

A.3 Filing-frequency pilot tests

The following tables come from the pilot tests described in Section 2.2. They cover 30 firms (financial data) and 15 firms (text) and are illustrations, not estimates for the panel.

Table A.5: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.6: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	∼52% substantive
Risk Factors content	Median ∼30,000 words	Median ∼9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	∼11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
Item 1A length (H1)	Feasible (annual word count)	Limited (52% substantive)
Item 1A risk-word density (H2)	Feasible (annual LM ratio)	Limited (52% substantive)
Literature alignment	Standard	Uncommon
Expected observations	∼6,400 firm-years	∼19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required in Form 10-Q (Part II, Item 1A) only for "material changes" from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.5, 2.1, and 2.2 for detailed results.

Table A.7: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: "MD&A Found" and "Risk Subst." count filings with >200 substantive words. "Risk Boilerplate" counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.4 Parser implementation notes

The parser operates on the HTML DOM before flattening the document into text, so that heading structure and placement survive the conversion. Candidate headers are detected only when they match an "Item X" pattern, are short, carry a heading cue, and are not hyperlinks. After filtering table-of-contents entries and reference phrases, the parser keeps the candidate followed by the longest section and then bounds the section by the next numbered item header.

```
def find_candidates(node):
    for el in walk_dom(node):
        text = clean_text(el)
        if not re.search(r"\bItem\s+[1-9][A-C]?\b", text, re.I):
            continue
        if word_count(text) > 25:
            continue
        if not has_heading_cue(el):
            continue
        if has_hyperlink(el):
            continue
        yield el


def select_item(doc, item):
    candidates = []
    for cand in find_candidates(doc):
        nearby = nearby_text(cand, radius=300)
```

```
        if item == "1A" and "risk" in nearby.lower():
            candidates.append(cand)
        elif item == "7" and any(k in nearby.lower() for k in [
            "management", "discussion", "md&a"
        ]):
            candidates.append(cand)

    candidates = [
        c for c in candidates
        if not is_table_of_contents(c)
        and not is_reference_phrase_before(c)
    ]
    return choose_candidate_followed_by_longest_section(candidates)
```

When no candidate section of at least 200 words is found, the same logic is applied to the referenced exhibit. In that case, numbered item headers are replaced by title-based patterns such as "Risk Factors" or "Management's Discussion and Analysis", because exhibits do not contain item numbers.

A.5 Additional robustness tables

Table A.8: Robustness of the H1/H2 decomposition (365-day window)

	$\ln(\text{Words}^{1A})$	UncDensity	LitDensity	N
Baseline (Table 4.3, col. 5)	+0.084*** (+8.64)	−0.106 (−0.11)	−4.709*** (−5.38)	5,782
+ market volatility in window	+0.085*** (+8.47)	−0.189 (−0.21)	−4.641*** (−5.32)	5,782
Volatility without earnings-release days	+0.069*** (+7.07)	−0.426 (−0.44)	−4.087*** (−4.53)	5,782
Idiosyncratic volatility as DV	+0.107*** (+8.88)	−2.468** (−2.17)	−5.227*** (−4.72)	5,782
Filing-month × year FE	+0.077*** (+8.36)	+0.043 (+0.05)	−4.046*** (−5.42)	5,782
Calendar year-month of filing FE (instead of fiscal year)	+0.078*** (+8.52)	+0.064 (+0.07)	−3.958*** (−5.20)	5,782
Firms with a December fiscal-year end only	+0.079*** (+8.12)	+0.677 (+0.70)	−2.608** (−2.47)	4,200
SIC3 + year FE	+0.080*** (+7.79)	−0.299 (−0.28)	−4.138*** (−3.93)	5,782
FF48 + year FE	+0.083*** (+9.02)	−0.020 (−0.02)	−4.958*** (−5.70)	5,782
Today's SIC2 + year FE	+0.086*** (+9.48)	−0.166 (−0.18)	−4.500*** (−5.28)	5,782
Excluding utilities (49) and health services (80)	+0.085*** (+7.59)	−0.713 (−0.73)	−4.421*** (−4.72)	5,112
Excluding weighted-average share counts	+0.080*** (+7.30)	−0.366 (−0.37)	−4.858*** (−4.91)	5,392
Excluding imputed balance-sheet values	+0.087*** (+6.27)	−1.494 (−1.30)	−4.332*** (−3.37)	3,820
Balanced panel, FY 2012–2024	+0.097*** (+8.25)	+1.105 (+0.89)	−4.807*** (−4.08)	3,367
Firms with ≥ 3 years	+0.084*** (+8.58)	−0.134 (−0.14)	−4.731*** (−5.42)	5,765
Firms with ≥ 5 years	+0.084*** (+8.27)	−0.168 (−0.18)	−4.699*** (−5.26)	5,613
Firms with ≥ 7 years	+0.082*** (+8.05)	−0.203 (−0.20)	−4.598*** (−4.98)	5,432
+ lagged dependent variable	+0.062*** (+7.42)	−0.181 (−0.27)	−3.495*** (−4.94)	5,780
SE clustered by firm only	+0.084*** (+8.31)	−0.106 (−0.10)	−4.709*** (−4.71)	5,782
SE clustered by year only	+0.084*** (+14.10)	−0.106 (−0.17)	−4.709*** (−9.12)	5,782
Counts: $\ln(\text{Words})$, $\ln(1+n^{\text{unc}})$, $\ln(1+n^{\text{lit}})$	−0.079*** (−4.80)	+0.164*** (+7.37)	−0.015 (−1.25)	5,855
Text of year $t-1$ instead of t	+0.080*** (+7.74)	+0.117 (+0.13)	−4.103*** (−4.92)	5,345
Level + change: Level $t-1$	+0.085*** (+8.24)	+0.107 (+0.11)	−4.340*** (−4.79)	5,289
Level + change: Change $t-1 \to t$	+0.067*** (+2.69)	−0.949 (−0.46)	−2.456 (−1.05)	5,289
Post-2019 interaction: FY $\leq$ 2019 (main effect)	+0.067*** (+5.19)	−1.597 (−1.51)	−4.886*** (−4.76)	5,782
Post-2019 interaction: × FY $\geq$ 2020	+0.052** (+2.43)	+4.418** (+2.40)	−0.164 (−0.12)	5,782

Notes. Each row is a separate regression of the column-(5) specification with the change described; common controls, industry (historical SIC2) and year FE and two-way clustered standard errors unless stated. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.9: Industries in the $\geq$ 20-peer H3 sample

SIC2	Firm-years	Firms	Share	Share of full H3 sample
73	754	83	0.231	0.164
49	527	46	0.162	0.113
28	486	48	0.149	0.108
35	455	44	0.139	0.102
38	431	40	0.132	0.095
36	407	39	0.125	0.091
37	202	21	0.062	0.053

Notes. SIC2 = two-digit SIC code at filing (e.g. 28 chemicals and pharmaceuticals, 35 industrial machinery and computers, 36 electronic equipment, 38 instruments, 49 utilities, 73 business services).

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction audit 1 (Section 2.4.4)

Tool. Google Gemini 2.5 Flash, accessed through the OpenAI-compatible Gemini API, temperature 0.

Input and task. The model received one stored section at a time (the full extracted text of Item 1A or Item 7 with its word count, ticker and fiscal year) and a five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) with three levels (Pass, Minor, Fail). It did not receive the rest of the filing, so boundary judgements are made from the extracted text alone. The prompt is part of `scripts/validate_extraction.py`.

Sample. 200 sections, drawn at random within five three-year fiscal-year strata (2010–2012, . . . , 2022–2024), 20 per stratum and section type; Item 1A and Item 7 were drawn independently (197 distinct filings). The texts the model rated are identical to the stored texts used in the analysis.

Extraction audit 2

Tool. Claude Opus 5.5 (Anthropic), used through the coding assistant described below, on 1 November 2026.

Input and task. The same 200 sections and the same rubric, blind to the ratings of audit 1. Because of the length of the sections, the model received for each section the first and last 1,200 characters, a 600-character excerpt from the middle and the word count. It therefore judges the start and end boundaries on the same evidence as audit 1 but sees less of the body of each section. The ratings are stored in the repository (`admin/claude_audit/`).

Results. Table B.1 reports both audits. The two models agree on the overall verdict for 96.5% of the sections.

Table B.1: Extraction audits: sections rated acceptable ($n = 200$ sections, 100 Item 1A and 100 Item 7)

	Gemini 2.5 Flash (full section)			Claude Opus 5.5 (excerpts)	
Criterion	Strict %	Lenient %	95 % CI	Lenient %	Agreement
Correct section	96.0	96.0	[92.3, 98.0]	97.0	99.0
Start boundary	95.5	96.5	[93.0, 98.3]	96.5	100.0
End boundary	96.0	96.5	[93.0, 98.3]	96.0	98.5
Content quality	95.5	97.5	[94.3, 98.9]	97.5	100.0
Completeness	94.5	95.5	[91.7, 97.6]	97.5	98.0
Overall (all five)	89.5	93.5	[89.2, 96.2]	95.0	96.5
Gemini: sections with at least one Fail, by fiscal-year stratum					
FY 2010–2012	5 of 40				
FY 2013–2015	3 of 40				
FY 2016–2018	2 of 40				
FY 2019–2021	2 of 40				
FY 2022–2024	1 of 40				
Manual check by the author (51 sections, correct section and both boundaries)					
Correct section	0 deficient; agrees with Gemini 94.1%			agrees with Claude 98.0%	
Start boundary	2 deficient; agrees with Gemini 100.0%			agrees with Claude 100.0%	
End boundary	2 deficient; agrees with Gemini 100.0%			agrees with Claude 98.0%	
All three	4 deficient; agrees with Gemini 100.0%			agrees with Claude 98.0%	

Notes. Strict = rated Pass; lenient = Pass or Minor issue. The sample is stratified into five three-year fiscal-year strata with 20 sections per stratum and section type; Item 1A and Item 7 were drawn independently (197 distinct filings). 5 sections with no extracted text are counted as Fail on every criterion. Gemini saw the whole extracted section; Claude saw its first and last 1,200 and a middle 600 characters and the word count, blind to the Gemini verdicts. Neither saw the surrounding filing. Agreement is the share of sections on which both audits give the same verdict, acceptable or Fail. Wilson score intervals. Both audits fail the same 8 sections on the overall criterion; 5 fail only under Gemini and 2 only under Claude. Manual check: error rate on the 50 random sections 6.0% (Wilson 2.1–16.2%), against the original filings, blind to both audits.

Limitations. Both audits are LLM judgements of the extracted text without the original filing, and the second model is used as a check of the first, not as ground truth. The author checked 51 sections by hand against the filings on EDGAR, blind to both audits (50 random sections and the remaining LLM-flagged section; labels in `admin/human_labels_template.csv`). The manual verdict on correct section, start and end boundary agrees with audit 1 on all 51 sections and with audit 2 on 50; the four sections the author marked as deficient are the four that audit 1 failed on these criteria. The audits do not compare the parser with other extraction methods, so no claim is made about its accuracy relative to plain-text or XBRL-based extraction.

Code development, analysis and text revision

Tools. AI code completion in VS Code during the original development; for the revision of the thesis, an AI coding agent in VS Code (GitHub Copilot with Claude Opus 5.5).

Purpose. The agent was used to rewrite and test the data and estimation scripts, to re-run the analysis, to generate the tables and figures from the stored results, to check the references against the Crossref database, and to draft revised text of Chapters 2 to 5, the appendices and the conclusion.

Contribution and verification. The research questions, the hypotheses, the decisions on the revision and the decision rule for H3 are the author's. All code, results and text produced with the agent were reviewed by the author; every number in the text is generated by the scripts in the repository, and a script checks the numbers in the text against the generated results.

Writing assistance

Tool. AI writing assistant integrated in VS Code.

Purpose. Grammar and style suggestions for English prose.

Verification. Every suggestion was reviewed before acceptance.